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NeRF for metaverse: a comprehensive review of neural radiance field-based techniques for digital realm synthesis

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Abstract
In the realms of next-generation internet, the metaverse aims to offer an entirely immersive and self-sustaining virtual environment for diverse activities and interactions. Recent advancements in blockchain technology, extended reality (XR) and artificial intelligence (AI) have paved the way for the metaverse to become an imminent reality. However, a longstanding challenge has been synthesizing novel views of intricate scenes within the simulated environment from a constrained set of input images. The advent of neural radiance field (NeRF) has significantly eased the process of rendering photorealistic views of scenes with intricate geometry, achieved by optimizing a continuous volumetric scene function. This study offers a comprehensive review and analysis of pioneering NeRF advancements in the context of the metaverse. Additionally, we introduce a novel taxonomy for the categorization of NeRF mechanisms, primarily based on their application areas and conduct an in-depth examination of architectural modifications in NeRF models. We also elucidate performance protocols, including commonly employed benchmark datasets and evaluation metrics for assessing these algorithms. Despite the existing contributions in the literature on view synthesis, numerous open research problems still persist, warranting the attention of researchers in this active field of study.

Keywords Metaverse · View synthesis · NeRF · Volume rendering · Digital clone

1 Introduction

The term "Metaverse", originally introduced by Neal Stephenson in his science fiction work "Snow Crash [1]", merges the prefixes "meta" and "verse" to denote a domain that transcends the conventional universe. This concept emerges from the integration of virtual reality (VR) and augmented reality (AR), creating a comprehensive digital environment. The popularity of the metaverse has grown during the COVID-19 pandemic, as more people

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turned to online platforms for work, entertainment, and social interaction. Within this virtual world, users can buy and sell virtual assets using cryptocurrency, meet friends, explore virtual spaces, purchase goods, and attend events while representing themselves as customizable digital avatars, often referred to as digital clones. Tech giants such as Meta (formerly Facebook), Microsoft, Tencent, and NVIDIA [2] have announced their ventures into the metaverse, indicating the growing global attention and interest in this paradigm.

In this evolving landscape, digital realm synthesis has emerged as a central challenge and opportunity. It refers to the process of creating high-fidelity, interactive, and immersive digital counterparts of physical environments, objects, and entities. This synthesis is vital for enabling realistic interaction, navigation, and communication within the metaverse, and is powered by advances in neural rendering, machine learning, and 3D modeling. The evolution of the metaverse unfolds through initial steps of constructing digital replicas of real-world objects into the virtual environment, followed by facilitating user engagement as digital inhabitants, and ultimately achieving a seamless trustworthy integration of virtual and physical realms in surreality. Attaining this requires advancements in digital twin technology, AI-enabled solutions for behavioral realism, and decentralized systems like blockchain for secure and transparent transactions. The field of computer vision has seen a promising direction in encoding objects and scenes using multi-layer perceptron (MLP) models [3], which map 3D spatial locations to implicit representations like signed distance functions. However, these approaches encounter difficulties in accurately rendering realistic environments with intricate geometries when compared to methods that utilize discrete representations like triangle meshes or voxel grids. Figure 1 depicts the evolution of view synthesis mechanisms [4–16] over the past decades. A persistent obstacle in the generation of novel views within the metaverse and synthetic environments is the optimization of continuous volumetric scene functions with limited input views. NeRFs have emerged as indispensable elements in metaverse and AR exploration, offering proficient and inventive rendering capabilities. The initial release of NeRF [17] for novel view synthesis (NVS) was introduced in 2020, and subsequent advancements in its domain have found applications in various fields such as facial modeling, entertainment, gaming, architectural visualization, medical imaging, autonomous systems, and etc.

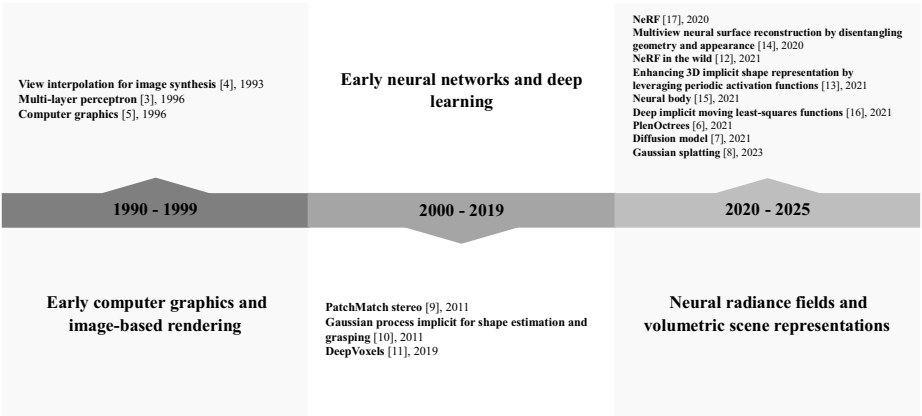


Fig. 1 A timeline illustrating the evolution and advancements in view synthesis

The literature demonstrates significant growth in this dynamic research field, with pioneering contributions from various investigators. Zhu et al. [18] conducted a comprehensive review of recent NeRF advancements, covering many essential domains. However, it uncovers latest developments in synthesized view generation approaches emerging up till 2023. Tewari et al. [19] provides a thorough analysis of NeRF, focusing on its fundamentals and application areas. Yet, it may be analyzed that certain significant aspects pertaining to newer applications such as automated systems are overlooked in this study. Furthermore, Table 1 covers the key contributions of our research over state-of-the-art (SOTA) review papers. Though broader domain of radiance field application are covered in these studies many of the recent advancements in the sub-domains of scene recognition, sketch or text to photorealistic face generation, hand-object interaction, robotics, medical imaging, and etc. are overlooked.

Given that the majority of recent approaches hinge on two vital characteristics – the robustness of NeRF and its applications – a more in-depth examination of the SOTA is necessary. This article presents several distinctive contributions as listed below:

i. We propose a novel taxonomy for classifying NeRF mechanisms based on their application areas. This taxonomy provides a structured framework that helps researchers clearly delineate the diverse domains explored using radiance fields.
ii. We conduct a detailed comparative analysis of SOTA NeRF models, examining their underlying concepts, deployed datasets, performance metrics, and the limitations of existing approaches.
iii. We provide a comprehensive review of NeRF architectural modifications, categorizing them based on their impact on energy efficiency, rendering speed, training data requirements, and the generation of high-quality novel views.
iv. We present a structured overview of performance evaluation protocols, outlining commonly used benchmark datasets and key metrics for assessing NeRF algorithms.
v. Finally, we identify key open research challenges uncovered in this study and propose potential solutions to drive future advancements in the field.

Table 1 Key contributions of our work compared to other survey papers. *Abbreviations*: CG – Computer graphics, IM – Interactive media development, AS – Automated systems, VS – Visual computing, BD – Benchmark datasets, PM – Performance metrics

Survey works	Scope and coverage	Taxonomy	Domain-specific NeRF applications				NeRF architectural modification	Evaluation protocols		Open research challenges
			CG	IM	AS	VC		BD	PM	
Deep review & analysis of recent NeRFs [18]	×	×	✓	✓	✓	✓	×	×	×	✓
Advances in neural rendering [19]	×	×	✓	✓	×	✓	✓	×	×	✓
NeRF in 3D vision [20]	×	✓	✓	✓	✓	×	✓	✓	✓	✓
NeRF for Metaverse (Ours)	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓

Fig. 2 A visual schematic depicting the overall structure of the article

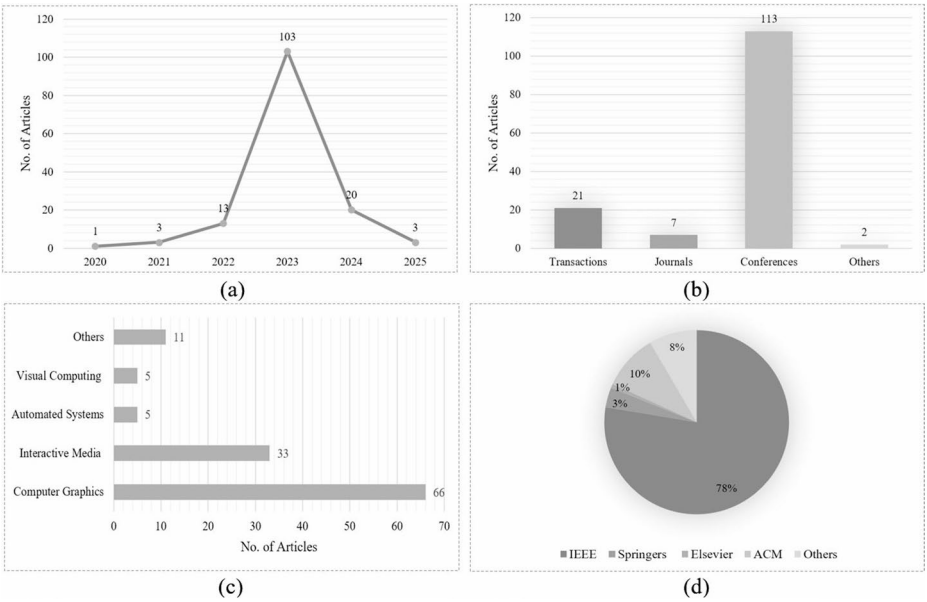


Fig. 3 Overall distribution of NeRF literature (**a**) Yearly distributions of articles (**b**) Categorization of articles by type (**c**) Application areas (**d**) Publisher wise distribution

The organization of the article is delineated in Fig. 2. Section 2 defines the study's scope and coverage. Section 3 delves into a comprehensive review and analysis of NeRF, focusing on its application domains and architectural enhancements. Section 4 addresses the evaluation methodologies, detailing the datasets and metrics utilized to assess the performance of diverse NeRF techniques tailored to specific applications. Section 5 highlights the unresolved research challenges associated with NeRF-based approaches and proposes potential avenues for future exploration. Finally, Sect. 6 synthesizes the key findings of the study and concludes the article.

2 Scope and coverage

The primary objective of this research is to perform an in-depth examination of SOTA methodologies, tracing their development from inception through 2025. To achieve this, we systematically examined over 200 articles sourced from reputable repositories such as IEEE Xplore, Scopus, Web of Science, ResearchGate, among others. Following an extensive review of the literature, we carefully selected a total of 174 articles, out of which 143 are research articles chosen for comparative analysis and included in the study. Our survey focuses on the latest literature, as evidenced by Fig. 3a, which illustrates that the major-

ity of the selected articles are published between 2022 and 2025. Furthermore, our analysis revealed that the pioneering contributions in radiance field mechanisms predominantly appear in top-rated conferences, as highlighted in Fig. 3b, which illustrates the distribution of article types in our study. Figure 3c indicates that the majority of the reviewed NeRF approaches are applied in two main areas: computer graphics and interactive media development. Additionally, the distribution of selected articles based on publishers is presented in Fig. 3d.

3 Neural radiance fields

In the preceding section, we delved into the recent advancements in synthesizing novel views of object or scene from early computer graphics to NeRF. Radiance field-based techniques have simplified the rendering of novel perspectives of complex scenes by utilizing a restricted number of input views. Short after the first release of NeRF a major advancement has been seen for deploying this technology in various areas of application. Therefore, in this section, we explore the significant advancements in radiance field-based applications and provide an in-depth analysis of NeRF's architectural modifications through hardware and software integrations.

3.1 Domain-specific NeRF applications

The metaverse represents a comprehensive and immersive digital universe that integrates various interconnected mixed-reality environments. Rather than being confined to a single platform, it encompasses multiple interlinked simulated realms, forming a cohesive, persistent, and collective digital experience. One key area within the metaverse is the 3D synthesis of real-world objects using radiance field-based methods. These 3D reconstructions can be deployed across various subdomains within the metaverse. To systematically classify these applications, we have introduced a novel taxonomy, as illustrated in Fig. 4, which categorizes NeRF-based applications into five distinct groups: computer graphics, interactive media development, automated systems, visual computing, and other emerging domains. This taxonomy not only captures the breadth of NeRF's use cases but also reflects the field's evolution from foundational techniques to more complex, real-world applications.

The diverse applications highlighted in the taxonomy exhibit a clear progression from passive scene understanding and rendering toward interactive, intelligent, and cross-modal systems. Initially, NeRF techniques focused on scene reconstruction and view synthesis, forming the backbone of applications in architectural rendering and virtual reality. As methods matured, the field branched into character modeling, particularly face and body modeling, enabling dynamic avatars for the metaverse and prosthetic design. These developments laid the groundwork for interactive media applications, including video game design and multimedia editing, where NeRFs enable inpainting, style transfer, and text-guided scene manipulation. Parallel to this, NeRF began penetrating automated systems, such as robotics and autonomous navigation, driven by the need for 3D perception in real-world environments. A more recent and promising direction is visual computing and medical imaging, where NeRF's high-fidelity modeling enables organ reconstruction or signal interpretation. This chronological and functional progression reflects an overarching trend from passive

NeRF	***Computer graphics***	***Scene understanding***	***Scene reconstruction*** • *2D image-based reconstruction* • *3D video-based reconstruction* ***Scene recognition***
		Avatar and character	***Face modeling*** • *Expression modulation* • *Facial morphing and prosthetic addons* • *Sketch-to-photo realistic face generation* • *Text-driven facial animation* ***Body and hand modeling*** • *Manipulating body posture* • *Hand-object interaction*
		Rendering	***Architectural 3D rendering*** ***Augumented and virtual reality***
	Interactive media development	***Multimedia editing***	***Style transfer*** ***Image inpainting*** ***Scene relighting*** ***Text-guided editing*** ***Object detection and segmentation***
		Video game design	
	Automated systems	***Autonomous navigation***	
		Robotics	
	Visual computing	***Medical imaging***	
		Signal modeling	
	Others		

Fig. 4 Our proposed taxonomy of NeRF applications

scene modeling to interactive, intelligent, and multimodal understanding indicating NeRF's trajectory toward becoming a versatile foundation for real-time, embodied AI in both virtual and physical spaces. In the subsequent sections, we delve into the established NeRF methodologies categorized according to predefined types in the taxonomy. This exploration encompasses their fundamental principles, utilized datasets, and performance metrics.

3.1.1 Computer graphics

NeRFs have revolutionized computer graphics by enabling highly detailed and photorealistic scene reconstruction, avatar and character modeling, and architectural 3D rendering. By synthesizing realistic and novel views from 2D images or 3D video data, NeRFs significantly enhance visual quality and immersive experiences in AR/VR applications.

a **Scene understanding**
3D scene understanding refers to the process of reconstructing, recognizing, and interpreting three-dimensional environments from visual data. Figure 5 highlights recent advancements in NeRF applications that contribute to this field. It encompasses techniques such as scene reconstruction, object recognition, and depth estimation to generate detailed and structured 3D representations.

Scene reconstruction Scene reconstruction pertains to the method of developing intricate and lifelike digital models of both physical and virtual spaces within this expansive metaverse. NeRF leverages neural networks to model 3D scenes, capturing intricate details in radiance and lighting, thereby enhancing the realism and interactivity of metaverse experiences. These reconstructed scenes serve as the backdrop for users to interact, and engage within the metaverse, creating a seamless and immersive digital universe.

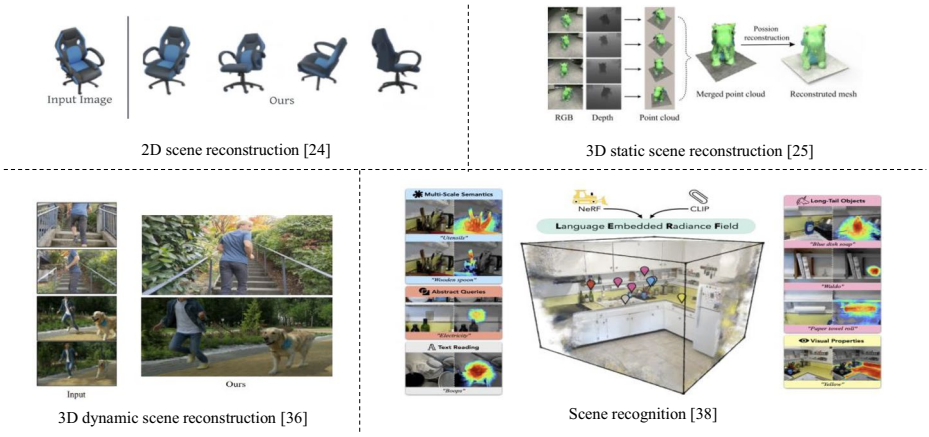


Fig. 5 Applications of radiance fields in scene understanding, including scene reconstruction and recognition

2D image-based reconstruction While NeRF is primarily designed for capturing and modeling 3D scenes from multiple input images or videos, its application in 2D image-based reconstruction focuses on generating detailed and realistic representations from a single 2D scene. This approach is particularly useful in virtual environments where depth information is limited. In 2023, Pavllo et al. [21] presented a comprehensive 2D image-to-3D reconstruction framework utilizing NeRF, which aims to reconstruct the shape, appearance, and pose of an object from a single image without relying on ground-truth pose information. This method utilizes a hybrid inversion scheme combined with principled pose estimation techniques to enhance processing speed. It achieves efficiency with as few as 10 inversion steps. However, existing limitations include the need for improved surface quality and further exploration of pose distribution inference from the available data. Deng et al. [22] put forward NeRDi, a single-view NeRF synthesis framework designed to leverage 2D priors from diffusion models, tackling the challenging issue of 2D-to-3D reconstruction. The framework refines NeRF representations by minimizing diffusion loss and incorporates language guidance to improve both semantic alignment and visual consistency. The method demonstrated the capability of synthesizing high-quality novel views, even in zero-shot scenarios, and introduced a geometric regularization term to address depth map ambiguity. However, the approach relied on pre-trained image models and may have exhibited biases, been less robust for highly deformable instances, and been more suitable for object-centric scenes. Also, Wimbauer et al. [23] addressed the challenge of inferring meaningful geometric scene representations from a single image using NeRFs. The deliberated approach introduced implicit density fields, which mapped volumetric density within the frustum of the input image, making the scene representation less complex compared to NeRFs. By employing direct color sampling from accessible viewpoints and leveraging self-supervised learning techniques on datasets, the approach effectively predicted accurate geometry in occluded areas. Further, Gu et al. [24] tackled the challenge of NVS from a single image by employing NerfDiff, a framework that combined a NeRF with a 3D-aware conditional diffusion model. While the method showed promising performance, it had certain limita-

tions such as the requirement for atleast two training views and computational cost during fine-tuning.

3D video-based reconstruction The initial release of NeRF [17], primarily focused on generating 3D novel views of objects and scenes from a single monocular input video. Subsequently, this domain has witnessed numerous advancements and developments. Yuan et al. [25] proposed a framework for high-quality view synthesis using sparse RGB-D data, allowing realistic rendering of novel views even with limited input. The stated technique involved constructing a geometric scene proxy from captured RGB-D images, pre-training a network with synthetic images, and fine-tuning it with real-world data. It also leveraged a patch discriminator and 3D color prior for more robust supervision and improved synthesis accuracy. Despite reducing the dependency on large input datasets, the given approach had limitations, particularly with large, complex outdoor environments or scenes containing missing elements, high specularity, thin structures, or furry textures. Moreover, the framework struggled to generate novel views not encompassed by the captured input without requiring extensive training data. Pan et al. [26] discussed a framework designed to generate high-quality novel view renderings of refractive objects with curved paths, refractive index reconstruction from silhouettes, and hierarchical path sampling techniques. However, promising results quantitatively and qualitatively are stated; still, challenges with real scenes' geometric details and foggy artifacts are noted. Hu et al. [27] developed Point2Pix, a point rendering system that integrates 3D sparse point clouds with 2D image pixels to generate high-quality, photorealistic images, particularly for indoor environments. This methodology introduces efficient ray sampling and multi-scale radiance fields, which enhance rendering speed and facilitate the extraction of distinctive 3D features. Experimental results on the ScanNet and ArkitScenes datasets validate its effectiveness; however, limitations include constraints on rendering time and challenges in generating images for arbitrary settings. Meanwhile, Huang et al. [28] propounded a framework called reference-guided super-resolution NeRF (RefSR-NeRF), which enhances NeRF's capability for super-resolution and photorealistic NVS. While NeRF has been widely recognized for its effectiveness in neural rendering, it typically faces challenges when producing high-resolution outputs, primarily due to its difficulty in capturing fine details and its computational demands. RefSR-NeRF overcomes these challenges by initially learning a low-resolution NeRF model and subsequently refining it with high-frequency details derived from a high-resolution reference image.

Jiang et al. [29] put forward AligNeRF, which combines convolutional layers and multilayer perceptron's to enhance the model's performance and address misalignment and high-frequency detail issues, resulting in improved image quality without significant additional computational costs. Zhu et al. [30] offered X-NeRF, an explicit approach utilizing 3D sparse generative convolutional neural networks (CNNs) to complete sparse point cloud tensors. This method generated explicit radiance fields from limited RGB-D input views, enabling fast volumetric rendering during inference. This innovative approach addressed a challenging problem in the context of NeRFs by exploring the possibility of training a single model to represent multiple scenes, even when provided with extremely sparse and non-overlapping input views. Phongthawee et al. [31] proposed NeX, a method for real-time NVS that employs multiplane images where each pixel is parameterized as a linear combi-

nation of spherical basis functions. These functions are learned by a neural network, allowing for efficient representation of view-dependent effects. The enhanced version, NeX360 [31], extends this approach to unbounded 360° scenes by incorporating knowledge distillation, which enables the fusion of multiple multiplane images (MPIs) from various viewing perspectives. The proposed study demonstrated improved performance on benchmark datasets and challenging view-dependent effects, highlighting limitations related to MPI artifacts and the inability to reproduce complex effects like light sparkles, refraction, and reflections from transparent materials.

Meanwhile, to address the limitations of rendering reflective objects, leading to blurriness and distortions Yin et al. [32] introduced a multi-space NeRF approach that models scenes by utilizing multiple parallel sub-spaces. The proposed method enhances the neural network's ability to represent reflective and refractive surfaces, resulting in improved accuracy and realism in high-quality scene rendering. Huang et al. [33] expounded the local implicit ray function (LIRF), a versatile neural rendering method designed for generating novel views through generalization. LIRF aggregated information from conical frustums to construct rays, allowing high-quality rendering at continuously-valued scales, and incorporated a transformer-based visibility weight estimation module to address occlusion challenges. The method shows comparable performance in NVS, despite some limitations related to view consistency, and computational cost, suggesting the potential benefit of introducing geometric priors in challenging scenes. Lee et al. [34] bring forward DP-NeRF, a NeRF-based framework aimed at reconstructing 3D scenes from blurred images while preserving both geometric and appearance consistency in the 3D space. DP-NeRF leverages a rigid blur kernel along with an adaptive weight proposal grounded in physical priors, enabling it to deliver high-quality visual results. The stated approach shows significant potential for extending its application to tackle other forms of image noise. Liu et al. [35] unveiled the concept of a semantic ray (S-Ray), a robust semantic field designed to learn from various scenes and effectively adapt to unfamiliar environments. The S-Ray integrates a cross-reprojection attention module, which consolidates extensive semantic data from multiple view reprojections, thereby improving its generalization capabilities. This advancement opens avenues for exploring complex vision tasks utilizing NeRF technology.

The advancements in 3D reconstruction methods extend beyond scene reconstruction from static input videos to also include scene reconstruction from dynamic inputs, where objects are in motion. Li et al. [36] tackled the problem of NVS in monocular videos of complex dynamic scenes by employing a volumetric image-based rendering framework DynIBaR, effectively addressing the limitations of dynamic NeRFs. Unlike traditional methods that encode scene dynamics within neural networks, DynIBaR models motion trajectory fields using learned basis functions, ensuring smooth motion reconstruction and temporal coherence across frames. This technique allows DynIBaR to handle long-duration videos with complex object motion and uncontrolled camera trajectories, making it highly effective for real-world applications. Additionally, DynIBaR introduces cross-time rendering, which enforces temporal consistency by adjusting rays across different timestamps, preventing flickering and improving realism. Its motion segmentation module accurately separates static and dynamic elements, enhancing rendering sharpness. With over 50% lower learned perceptual image patch similarity (LPIPS) errors compared to SOTA methods, DynIBaR delivers photo-realistic synthesis even in challenging environments, proving its superiority in dynamic view synthesis. However, it has limitations in handling small fast-moving

objects and maintaining multi-view consistency for all static content. Sabour et al. [37] addressed a key issue in training NeRF models, specifically dealing with distractors such as moving objects and transient photometric phenomena that disrupt scene reconstruction. And proposed a simple yet effective technique, RobustNeRF, which employed robust estimation principles to identify and mitigate the impact of distractors during optimization, leading to improved performance on various datasets. Unlike conventional NeRF models that assume photometric consistency, RobustNeRF models distractors as outliers in an optimization problem, using techniques like trimmed least squares (TLS) and iteratively reweighted least squares (IRLS) to dynamically adjust pixel weights. This ensures that static scene elements are reconstructed with high fidelity while transient artifacts are ignored, leading to cleaner and more accurate NVS. Additionally, RobustNeRF employs spatially-aware filtering, leveraging local patch-based analysis to distinguish structured scene elements from random transient noise. This approach significantly improves scene fidelity, achieving higher peak-signal-to-noise ratio (PSNR) and structural similarity index measure (SSIM) scores while reducing LPIPS errors, meaning the synthesized images are closer to ground truth in perceptual quality. Its ability to automatically remove transient objects makes it a powerful tool for dynamic scene rendering, ensuring photo-realistic synthesis even in uncontrolled real-world scenarios. While RobustNeRF offered significant benefits in managing distractors, the method had some statistical inefficiencies on clean data.

Scene recognition It refers to the concept of utilizing the NeRF model in the domain analysing the given input scene. Kerr et al. [38] applied language-embedded radiance fields (LERFs), which integrate language embeddings from models such as contrastive language-image pretraining (CLIP) into NeRF for addressing open-ended 3D language queries. The suggested approach enabled the real-time generation of 3D relevance maps based on various language prompts, revealing promising applications in fields like robotics, vision-language integration, and interactions within 3D environments. However, it faced challenges in accurately capturing spatial relationships and depended heavily on calibrated camera matrices as well as the quality of multi-view data. Furthermore, Liu et al. [39] developed an innovative approach for the unsupervised continual adaptation of semantic segmentation networks across a range of new scenes through the application of neural rendering techniques. The approach improved adaptation performance by facilitating the transfer of knowledge between 2D and 3D representations. The NeRF-based replay mechanism effectively addressed the issue of catastrophic forgetting, allowing for the rendering of a virtually unlimited number of images for adaptation while maintaining a consistent storage cost. Meanwhile, Xue et al. [40] worked on the 3D-IntPhys framework, which learns intuitive physics from unlabeled visual data. This framework demonstrated proficiency in managing complex scenes that include fluids, rigid bodies, and granular materials, while also exhibiting strong generalization to previously unencountered scenarios. Nonetheless, the model's reliance on object masks poses challenges for application in intricate real-world environments. This suggests a future research direction focused on joint learning of segmentation, material properties, and dynamics from observational data.

Summary and trend analysis NeRF-based scene understanding has become a pivotal area in neural rendering, extending its role from NVS to the interpretation and reconstruction of complex environments, which is essential for immersive metaverse experiences. This

includes not only generating photorealistic 3D scenes from sparse 2D inputs but also recognizing scene semantics and spatial context. As summarized in Table 2, the field shows a concentration of research in static 3D scene reconstruction [17], covering tasks like NVS, handling reflections, and deblurring, while dynamic scene modeling [36] remains underexplored due to NeRF's inherent static nature. Current advances are addressing this limitation using temporal coherence, motion priors, and hybrid neural-geometric models. In contrast, 2D reconstruction [21] is the least explored area but is showing promise through the use of generative models like GANs and diffusion networks to enhance image fidelity from minimal inputs. Similarly, scene recognition [38] is gaining traction by integrating radiance fields with language embeddings to enable semantic understanding of complex scenes. Overall, trends indicate a growing shift toward multi-modal learning, generative modeling, and real-time scene adaptation for dynamic and semantically rich environments.

b Avatar and character

NeRF's scope has expanded from 3D scene rendering to encompass the field of avatar and character modeling. This adaptation facilitates the generation of intricately detailed and lifelike digital human clones, thereby enhancing digital visual representation within virtual environments. The process entails not only capturing and encoding the structural attributes of avatars but also encompasses their visual appearance and finer details, including clothing, facial features, and expressions. Leveraging NeRF has the potential to revolutionize the interactions and representations of virtual individuals in diverse digital contexts, delivering immersive and visually captivating experiences through the utilization of neural radiance for generating realistic avatars and characters. Figure 6 showcases recent advancements in NeRF applications for avatar and character modeling. Furthermore, this section incorporates an analysis of SOTA methods for modeling human face and body structures via NeRF models.

Face modeling Athar et al. [46] bring forward RigNeRF, a radiance field designed for precise control of human head movements while simultaneously modeling the entire 3D environment. Despite its impressive performance, RigNeRF had certain constraints: it is tailored to specific subjects, requiring separate training for each individual scene. To adequately capture a range of facial expressions and head poses, the method necessitated a training sequence of about 40–70s. Additionally, as with other NeRF-based approaches, the accuracy of camera registration directly influenced the quality of the output. Although RigNeRF enabled photorealistic facial reanimation, it posed potential societal risks if misapplied. In a related research, Yin et al. [47] developed a generalized method for inverting NeRF-GANs, allowing for the creation of high-quality, identity-consistent 3D animations of real-world subjects using only a single image. The suggested approach enabled the synthesis of new perspectives and facial expressions from real images. Kirschstein et al. [48] expounded a dynamic head reconstruction technique that integrates a deformation field with multi-resolution hash encoding ensembles. This methodology facilitates high-fidelity NVS from a sparse camera arrangement, enabling a comprehensive representation of scenes with intricate motion patterns. Additionally, the research introduced a high-frame-rate, high-resolution multi-view video dataset that includes over 4,700 sequences featuring more than 220 individuals, specifically tailored for the dynamic NVS of human heads. Bai et al. [49] offered a method

Table 2 A structured analysis of SOTA NeRF-based mechanisms for scene understanding. *Abbreviations*: S –Static, D – Dynamic

Year	Author	Reconstruction			Recognition	Key concept	Dataset	Performance
		2D	3D					
			S	D				
2020	Mildenhall et al. [17]	×	✓	×	×	3D novel views of objects and scenes via NeRF models via continuous volumetric function	Diffuse synthetic 360°	PSNR↑ = 40.15, SSIM↑ = 0.991, LPIPS↓ = 0.023
							Realistic synthetic 360°	PSNR↑ = 31.01, SSIM↑ = 0.947, LPIPS↓ = 0.081
							Real forward facing	PSNR↑ = 26.50, SSIM↑ = 0.811, LPIPS↓ = 0.250
2022	Yuan et al. [25]	×	✓	×	×	Leveraging sparse RGB-D images to enhance NeRF for high-quality view synthesis with improved depth awareness	Self-collected dataset	PSNR↑ = 25.27, SSIM↑ = 0.579, LPIPS↓ = 0.298
2022	Pan et al. [26]	×	✓	×	×	Renders novel views of both synthetic and real scenes containing refractive objects	Rendered real scenes (Pen)	PSNR↑ = 18.95, SSIM↑ = 0.494, LPIPS↓ = 0.315
2023	Pavllo et al. [21]	✓	×	×	×	Reconstructs shape, pose, and appearance from a single image by leveraging bootstrapped radiance field inversion	Real datasets following CMR (N=0)	FID↓ = 28.15
2023	Deng et al. [22]	✓	×	×	×	A single-view NeRF synthesis framework with general image priors from 2D diffusion models	Multi-view stereo (DTU MVS) dataset	PSNR↑ = 14.47, SSIM↑ = 0.465, LPIPS↓ = 0.421
2023	Wimbauer et al. [23]	✓	×	×	×	Explore density fields to enable high-quality 3D reconstruction from a single-view image by estimating scene geometry and appearance	KITTI dataset	PSNR↑ = 20.10, SSIM↑ = 0.764, LPIPS↓ = 0.144
							RealEstate10K	PSNR↑ = 24.00, SSIM↑ = 0.755, LPIPS↓ = 0.194

Table 2 (continued)

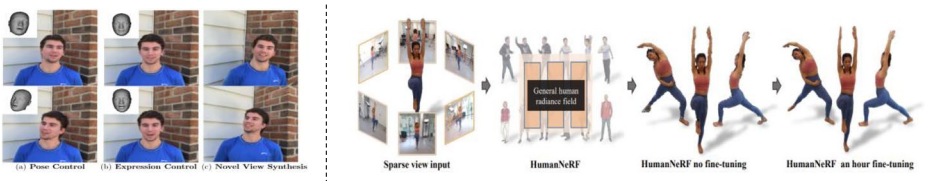
Year	Author	Reconstruction			Recognition	Key concept	Dataset	Performance
		2D	3D					
			S	D				
2023	Gu et al. [24]	✓	×	×	×	NeRF-guided distillation from 3D-aware diffusion models to achieve high-quality single-image view synthesis	SRN-ShapeNet (Cars)	PSNR↑ = 23.76, SSIM↑ = 0.920, LPIPS↓ = 0.076
							Amazon-Berkeley Objects	PSNR↑ = 32.84, SSIM↑ = 0.970, LPIPS↓ = 0.042
							Clevr3D	PSNR↑ = 34.81, SSIM↑ = 0.970, LPIPS↓ = 0.040
2023	Hu et al. [27]	×	✓	×	×	NeRF to achieve photorealistic rendering of point clouds, enhancing visual quality and scene realism	ScanNet (No fine-tuning)	PSNR↑ = 18.47, SSIM↑ = 0.723, LPIPS↓ = 0.484
							ArkitScenes (No fine-tuning)	PSNR↑ = 18.84, SSIM↑ = 0.734, LPIPS↓ = 0.471
2023	Huang et al. [28]	×	✓	×	×	Lightweight RefSR model to learn the inverse degradation process from NeRF renderings to target HR ones	Real forward-facing dataset (NeRF-LLFF×8)	PSNR↑ = 24.88, SSIM↑ = 0.850, LPIPS↓ = 0.466
2023	Jiang et al. [29]	×	✓	×	×	High-fidelity NeRF via alignment-aware training	Outdoor scenes from Mip-NeRF 360	PSNR↑ = 24.16, SSIM↑ = 0.678, LPIPS↓ = 0.327
2023	Zhu et al. [30]	×	✓	×	×	An explicit NeRF framework for reconstructing multi-scene 360° views from limited RGB-D inputs	Self-collected dataset using 7 RGB-D cameras to capture 6 seen and 4 novel scenes	PSNR↑ = 17.83, SSIM↑ = 0.584, LPIPS↓ = 0.508
2023	Phongthawee et al. [31]	×	✓	×	×	NVS based on enhancements of MPI that can reproduce view-dependent effects in real time	LF dataset	PSNR↑ = 27.05, SSIM↑ = 0.935, LPIPS↓ = 0.135
							Spaces dataset	PSNR↑ = 35.84, SSIM↑ = 0.985, LPIPS↓ = 0.083
2023	Yin et al. [32]	×	✓	×	×	It represents scenes using multiple parallel feature fields, enhancing neural network understanding of reflective and refractive objects	Self-collected dataset (Real)	PSNR↑ = 35.93, SSIM↑ = 0.948, LPIPS↓ = 0.066

Table 2 (continued)

Year	Author	Reconstruction			Recognition	Key concept	Dataset	Performance
		2D	3D					
			S	D				
2023	Huang et al. [33]	×	✓	×	×	LIRF enhances NeRF generalization by encoding spatially adaptive ray-based representations	LLFF dataset rendered at different scales ×4	PSNR↑ = 23.79, SSIM↑ = 0.760, LPIPS↓ = 0.373
2023	Lee et al. [34]	×	✓	×	×	Reconstructing 3D scenes from blurred images while preserving both geometric and appearance consistency in the 3D space	Real scene consisting of camera motion and defocus blur	PSNR↑ = 25.91 and 23.67, SSIM↑ = 0.775 and 0.729, LPIPS↓ = 0.160 and 0.108
2023	Liu et al. [35]	×	✓	×	×	Learning a generalizable semantic field with cross-reprojection attention	ScanNet dataset	PSNR↑ = 29.27, SSIM↑ = 0.865, LPIPS↓ = 0.127
2023	Li et al. [36]	×	×	✓	×	View synthesis for complex scenes with uncontrolled camera trajectories	Nvidia (Dynamic scene) UCSD (Dynamic scenes)	PSNR↑ = 30.91, SSIM↑ = 0.957, LPIPS↓ = 0.027 PSNR↑ = 36.47, SSIM↑ = 0.983, LPIPS↓ = 0.014
2023	Sabour et al. [37]	×	×	✓	×	RobustNeRF uses robust loss functions to ignore distractors and enhance rendering quality	Self-collected dataset natural (Statue) D^2NeRF synthetic scenes	PSNR↑ = 20.89, SSIM↑ = 0.750, LPIPS↓ = 0.280 PSNR↑ = 37.73, MS-SSIM↑ = 0.988, LPIPS↓ = 0.013
2023	Kerr et al. [38]	×	×	×	✓	Integrate language embeddings from models such as CLIP into NeRF for addressing open-ended 3D language queries	Real-world scenes	LERF = 80.30%
2023	Liu et al. [39]	×	×	×	✓	Unsupervised continual semantic adaptation through neural rendering	ScanNet dataset	Pseudo-label performance averaged = 45.00 ± 1.40

Table 2 (continued)

Year	Author	Reconstruction			Recognition	Key concept	Dataset	Performance
		2D	3D					
			S	D				
2023	Xue et al. [40]	×	×	×	✓	Learning 3D-grounded visual intuitive physics models from videos of complex scenes	FluidPour	SSIM↑ (InD/OoD) = 0.900/0.890
							FluidCubeShake	SSIM↑ (InD/OoD) = 0.940/0.930
							GranularPush	SSIM↑ (InD/OoD) = 0.890/0.690
2023	Seo et al. [41]	×	✓	×	×	NVS from sparse inputs by modeling a ray with a mixture density model	DTU (9-view)	PSNR↑ = 25.03, SSIM↑ = 0.879, LPIPS↓ = 0.065
							LLFF (9-view)	PSNR↑ = 25.20, SSIM↑ = 0.833, LPIPS↓ = 0.087
							Realistic synthetic 360° (8-view)	PSNR↑ = 18.99/23.84, SSIM↑ = 0.807/0.878, LPIPS↓ = 0.199/0.103
2024	Ren et al. [42]	×	×	✓	×	Exploiting uncertainty for distractor-free NeRFs in the wild	On-the-go dataset - High Occlusion (Patio-High)	PSNR↑ = 21.41, SSIM↑ = 0.718, LPIPS↓ = 0.235
							Robust-NeRF dataset (Android)	PSNR↑ = 23.50, SSIM↑ = 0.750, LPIPS↓ = 0.210
2024	Ko and Lee [43]	✓	×	×	×	Zero-shot 3D scene representation with invertible generative NeRFs	Real-image datasets (AFHQ 256^2)	FID↓ = 28.78
2024	Tang et al. [44]	×	✓	×	×	Given a NeRF of the original scene, and a single RGBD image of the transformed scene, enabling dynamic scene reconstruction and NVS	Objaverse dataset (NVS)	PSNR↑ = 25.9 ± 4.2, SSIM↑ = 0.924 ± 0.034, LPIPS↓ = 0.061 ± 0.040
2024	Li et al. [45]	×	✓	×	×	Unified learning framework that can combine NeRF and segmentation modules to perform context-aware 3D scene perception	ScanNet	PSNR↑ = $29.37_{2.8\uparrow}$, SSIM↑ = 0.919, LPIPS↓ = 0.110



Face modeling [46]

Body and hand modeling [65]

Fig. 6 A depiction of applying radiance fields in avatar and character modeling

for reconstructing facial avatars based on NeRF. The given technique utilizes a 3D-aware generative prior to tackle issues such as intricate facial dynamics and the absence of 3D data in monocular video sequences. The method proficiently learns a localized, low-dimensional subspace within the latent space of a 3D GAN. This enhancement improves the retention of unique facial characteristics while accommodating multiple driving signals, including RGB images, 3DMM coefficients, and audio inputs. Meanwhile, Shahreza and Marcel [50] put forward GaFaR, a method for reconstructing 3D faces using facial templates. The methodology employs a pretrained geometry-aware face generation network grounded in generative NeRF. It establishes a mapping between facial templates and the latent space of the generator network's intermediate representations. Ruzzi et al. [51] suggested GazeNeRF, a 3D-aware method for gaze redirection that overcomes the limitations of traditional 2D image manipulation by treating the facial region and eyeballs as distinct yet coordinated 3D structures. However, GazeNeRF is hindered by long training times, a common issue among NeRF-based models. To address this, Wang et al. [52] adopted the roll-out diffusion network (Rodin), which encodes a radiance field into a series of 2D feature maps and combines them into a unified 2D feature plane for efficient 3D-aware diffusion processing. Despite its advancements, Rodin still requires optimization to improve the speed of 3D diffusion sampling while leveraging abundant 2D data to mitigate the scarcity of 3D training datasets. Zhang et al. [53] presented the neural proto-face field (NPF), designed to disentangle identity, expression, and intricate details in 3D facial modeling. NPF utilizes diverse real-world image collections to refine its 3D generative model, employing a face prototype aggregation technique to manage variations in uncontrolled environments. It enhances personalization through new consistency loss functions.

Mendiratta et al. [54] emphasized the importance of capturing and editing complete head performances to produce photorealistic virtual characters, particularly for XR and media production applications. The suggested AvatarStudio approach enables editing of dynamic, full-head avatars using text-based prompts by integrating NeRF-driven dynamic head modeling with a text-to-image diffusion model. It also incorporates view-and-time-aware score distillation sampling for enhanced realism. Additionally, the approach utilizes a deformation network to propagate changes within a canonical space, greatly improving the realism and control of virtual head avatars. Hyung et al. [55] propounded the local editing NeRF, which incorporates three supplementary modules: the latent residual mapper, the attention field network, and the deformation network. This innovative approach facilitates localized 3D feature manipulation driven by textual inputs by estimating a 3D attention field. Notably, it operates independently of domain-specific labels, enabling broad applicability across diverse models and domains. Jo et al. [56] developed a comprehensive framework termed

conditional generative NeRF, which is proficient in generating multi-view images based on multimodal inputs, including both textual descriptions and images. Li et al. [57] divided the training procedure into two distinct stages: a multi-class 3D-aware GAN phase, followed by a 3D-aware translation phase. The initial phase utilized an innovative conditional architecture along with an efficient training methodology. In the subsequent phase, the well-optimized GAN architecture is employed to develop a 3D-aware translation system that maintained consistency across different views. To tackle issues related to view consistency, the model implemented several techniques, including a U-net-inspired adapter network, hierarchical representation constraints, and relative regularization loss. The generation of NeRFs from sketches poses challenges, such as the difficulty of translating 2D constraints into 3D forms and maintaining content richness from minimal sketches. To tackle these issues, Gao et al. [58] proposed SketchFaceNeRF, which creates and edits 3D facial NeRFs from sketches, producing high-quality images from multiple angles. The approach enhances single-view sketches by adding color and volumetric data, improving robustness against hand-drawn inputs. It also integrates a mask fusion module and a sketch-based optimization process to refine edits while preserving unaltered facial regions.

Subsequently, Guo et al. [59] introduced AD-NeRF, that uses audio features to condition a dynamic radiance field, enabling realistic talking-head video synthesis via volume rendering. The suggested approach captures head movements and facial expressions synchronized with the input speech. Shen et al. [60] propounded an end-to-end framework for talking head animation, utilizing a conditional NeRF to infer 3D structures from audio inputs. This approach effectively surpasses the limitations of traditional 3D morphable face models. The model is enhanced by incorporating motion data, creating an audio-motion dual-driven NeRF that yields more realistic animations. Additionally, the research proposed a spatially-adaptive dual-driven NeRF (SD-NeRF) with a cross-attention mechanism, allowing for precise control over facial movements and producing lifelike video outputs. Li et al. [61] adopted HiDe-NeRF, a model that enables high-fidelity, free-view talking head synthesis. This marks the first deformable NeRF model capable of one-shot learning and subject agnosticism. A multi-scale generalized appearance module and a lightweight expression-aware deformation module is also developed to enhance identity preservation and facial expression accuracy. Li et al. [62] contributed ER-NeRF, a streamlined architecture for real-time talking head synthesis that leverages a compact model structure. The proposed approach leverages spatial region contributions, a tri-plane hash representation, and a region attention module to enhance dynamic head reconstruction and audio-visual synchronization. This significantly improves both efficiency and output quality in talking head video generation.

Body and hand modeling In early 2021, Xu et al. [63] offered H-NeRF, a model aimed at 4D human rendering of moving subjects. The suggested framework utilized a structured implicit model of the human body along with geometric constraints to effectively integrate information from limited viewpoints. However, the model faced challenges when applied to scenarios that strayed from its assumptions of a complete body view, exhibiting sensitivity to estimated body shapes and image segmentation accuracy. Alongside, Su et al. [64] propounded A-NeRF based on an articulated radiance field to model human shape, appearance, and pose for high-quality, pose-dependent image synthesis. Zhao et al. [65] bring forward HumanNeRF, a neural representation engineered for high-quality free-view synthesis of

dynamic humans. HumanNeRF demonstrated improved generalization capabilities from limited RGB input streams, using an implicit blending technique to significantly enhance the generation of photorealistic human images in complex motion settings. Hu et al. [66] investigated hybrid volumetric-textural rendering (HVTR++), which leveraged driving signals (such as body poses) and view images to create realistic 3D human avatars, making it suitable for applications like telepresence.

Yu et al. [67] developed MonoHuman, a framework that renders consistent, high-fidelity human avatars under novel poses, allowing explicit control over poses using only monocular video supervision. MonoHuman achieved this through a shared bidirectional deformation module for consistent deformation and a forward correspondence search module to guide rendering based on features from keyframes. Despite its advantages, this method is limited by its reliance on accurate pose and mask annotations and necessitated case-specific training. Chen et al. [68] proposed generalizable model-based NeRF, aimed at synthesizing detailed novel view images of human performers from sparse multi-view data. This framework employed a geometry-guided attention mechanism to address challenges such as pose variations and self-occlusions, facilitating better alignment between geometric and pixel data. Despite its strong performance in both synthetic and real-world tests, it faced limitations with specific clothing styles, poses outside the training distribution, and artifacts in unobserved areas. Zhu et al. [69] devised CAT-NeRF, which tackled the challenge of rendering dynamic human body shapes with temporal appearance consistency in videos. CAT-NeRF employed a Tx2Former, which used two transformer layers to separate constant appearance features—representing general body shape—from frame-specific features that captured individual dynamics during sequential rendering. The approach incorporated a unique covariance loss to ensure accurate preservation of per-frame appearance, demonstrating effective performance on the H36M and ZJU-MoCap datasets. Peng et al. [70] presented IntrinsicNGP, an efficient method for NVS of human performers utilizing NeRFs. Unlike many existing methods that required extensive training, IntrinsicNGP achieved high-quality results within minutes using videos, by a continuous intrinsic coordinate representation and an optimized hash encoding module. The study also highlighted the potential applications of IntrinsicNGP for other dynamic objects, while emphasizing the ethical considerations related to generating synthetic human forms without consent, underlining the need for clear labeling of synthetic content.

Many models at the time are either hand-engineered or trained in controlled environments, limiting their applicability across diverse scenarios. Gao et al. [71] tackled the complex challenge of rendering new views and poses for previously unseen subjects using solely multi-view static images, without any video input. The given method combined a canonical NeRF with volume deformation, allowing for learning shared human features in a generalized space and connecting this space to input images for radiance and density predictions. Zhu et al. [72] propounded an approach to create 3D representations of clothed human avatars from multiple frames without needing specific camera calibrations or action restrictions, along with the release of the MVP-Human dataset. The proposed dataset included 400 individuals, each with 15 scans across different poses, captured from 8 views per pose, resulting in a total of 6,000 3D scans and 48,000 images. Wang et al. [73] offered a method for capturing dynamic clothed human performances from sparse-view or single-camera videos. The given double-layer NeRF framework separately tracked the motion of clothing

and the human body, improving accuracy and robustness in capturing dynamic clothing deformations. Additionally, it incorporated a physics-aware network to simulate realistic cloth dynamics and interactions with the body, making the approach effective across various garment types in real-world settings.

Summary and trend analysis NeRF has evolved beyond static 3D scene rendering to become a foundational technique for lifelike avatar and character modeling, enabling highly realistic digital human representations in virtual environments. This expansion encompasses detailed face, body, and hand reconstructions using radiance fields, capturing subtle features like facial expressions, clothing, and motion. Notable works include RigNeRF [46] for dynamic facial control, SketchFaceNeRF [58] for 3D synthesis from sketches, and AD-NeRF [59] for speech-synchronized talking heads. Techniques like AvatarStudio [54] and Local Editing NeRF [55] enhance user-driven customization via text prompts, while methods such as HumanNeRF [65], MonoHuman [67], and CAT-NeRF [69] address body motion and pose variability. These models leverage innovative architectures, including deformation fields, diffusion networks, and GAN-augmented pipelines, often trained with limited or sparse data, to achieve high-fidelity results. Despite breakthroughs, challenges like long training times, identity preservation, generalization across subjects, and ethical concerns around synthetic realism remain pivotal. Table 3 summarize representative models and comparative metrics across different modalities (face, body, and hand), highlighting ongoing advancements and future directions for NeRF-based avatar synthesis.

c Rendering

Rendering focuses on generating realistic or stylized images from pre-existing 3D models. Unlike scene reconstruction, which builds 3D models from 2D images or sparse data, rendering takes these models and synthesizes architectural views with enhanced visual quality. NeRF-based methods improve rendering fidelity for applications such as architectural visualization and AR/VR experiences. And generating immersive, high-fidelity visual experiences by rendering interactive 3D content for AR/VR applications. Figure 7 illustrates advancements in NeRF applications that contribute to the field of rendering. Furthermore, this section incorporates an analysis of SOTA methods for rendering.

Architectural 3D rendering Producing photorealistic representations of buildings and environments from structured 3D models, often used in design and visualization. Ye et al. [79] tackled the challenge of reconstructing 3D feature curves from calibrated multi-view images by developing a neural edge field (NEF). The approach optimized the NEF using a view-based rendering loss, with comparative performance on benchmark evaluations, which illustrates its capability to manage multi-modal information, though it faces limitations in processing textured objects. Erich et al. [80] proposed a comprehensive hardware and software framework for neural scanning of household objects using NeRF. This system facilitates object scanning by rotating the object while utilizing static cameras, demonstrating potential for achieving scanning quality comparable to that of commercial 3D scanning technologies.

Table 3 A comparative analysis of NeRF-based mechanisms in the domain of avatar and character modeling. *Abbreviations*: F – Face, B – Body, H – Hand

Year	Author(s)	F	B	H	Key concept	Dataset	Performance
2021	Guo et al. [59]	✓	×	×	The input audio features condition a dynamic NeRF, which is then volume-rendered to synthesize a high-fidelity talking-head video	Neural voice puppertry	SyncNet score = 5.895, AU error = 2.133
						Synthesizing Obama	SyncNet score = 5.411, AU error = 2.287
2021	Xu et al. [63]	×	✓	×	4D human reconstruction and rendering of moving subjects	Human 3.6 M	PSNR↑ = 25.01, SSIM↑ = 0.830, LPIPS↓ = 0.170
2021	Su et al. [64]	×	✓	×	A-NeRF, based on an articulated radiance field to model human shape, appearance, and pose	Human 3.6 M	Protocol I: PA-MPJPE↓ = 39.30
						MPI-INF-3DHP	PA-MPJPE↓ = 66.80
2022	Athar et al. [46]	✓	×	×	3DMM-guided deformable NeRF for full control of the human head along with simultaneously modeling the full 3D scene it is in	A smartphone-captured short video of a subject for training (Subject 1 on test data)	PSNR↑ = 29.55 LPIPS↓ = 0.136 FaceMSE↓ = 9.6e-5
2022	Yin et al. [47]	✓	×	×	A universal method to surgically fine-tune NeRF-GAN models in order to achieve high-fidelity animation of real subjects only by a single image	FFHQ (AniFace)	PSNR↑ = 24.92, FID↓ = 45.89
						CelebA-HQ (AniFace)	PSNR↑ = 25.61, FID↓ = 34.07
2022	Zhao et al. [65]	×	✓	×	Employs an aggregated pixel-alignment feature across multi-view inputs along with a pose embedded non-rigid deformation field for tackling dynamic motions	Self-collected sparse view input	PSNR↑ = 33.01, SSIM↑ = 0.984, LPIPS↓ = 0.033
2022	Gao et al. [71]	×	✓	×	A generalizable NeRF for novel view and pose synthesis using canonical space and volume deformation from multi-view images	Human3.6 M	NVS: PSNR↑ = 24.46, SSIM↑ = 0.910 NPS: PSNR↑ = 22.41, SSIM↑ = 0.881
						THuman dataset	NVS: PSNR↑ = 25.63, SSIM↑ = 0.935 NPS: PSNR↑ = 23.92, SSIM↑ = 0.911
2023	Kirschstein et al. [48]	✓	×	×	Dynamic head reconstruction technique that integrates a deformation field with multi-resolution hash encoding ensembles	Multi-view video dataset	PSNR↑ = 31.80, SSIM↑ = 0.875, LPIPS↓ = 0.212
2023	Bai et al. [49]	✓	×	×	3D-aware generative prior to tackle issues such as intricate facial dynamics and the absence of 3D data in monocular video sequences	FFHQ dataset	PSNR↑ = 32.57, SSIM↑ = 0.957, LPIPS↓ = 0.035

Table 3 (continued)

Year	Author(s)	F	B	H	Key concept	Dataset	Performance
2023	Shahreza and Marcel [50]	✓	×	×	Employs a pretrained geometry-aware face generation network grounded in generative NeRF	MOBIO (Swin)	TMR @ FMR of 10−3 = 98.98%
						LFW (Swin)	TMR @ FMR of 10−3 = 87.83%
2023	Ruzzi et al. [51]	✓	×	×	3D-aware method for gaze redirection	ETH-XGaze dataset	PSNR↑ = 15.45, SSIM↑ = 0.733, LPIPS↓ = 0.291
2023	Wang et al. [52]	✓	×	×	RODIN projects a 3D NeRF model into a unified 2D feature plane, enabling 3D-aware diffusion for enhanced scene generation	LAION-400 M	FID↓ = 26.10
2023	Zhang et al. [53]	✓	×	×	NPF for unsupervised robust 3D face modeling	CelebAMask-HQ	SSIM↑ = 0.921
2023	Mendiratta et al. [54]	✓	×	×	NeRF-based dynamic head representation with a text-to-image diffusion model and incorporating view-and-time-aware score distillation sampling (VT-SDS)	Self-collected dataset	The best in overall quality (Q4) 85.40% of the time
2023	Hyung et al. [55]	✓	×	×	Localized 3D feature manipulation driven by textual inputs by estimating a 3D attention field	FFHQ	PSNR↑ = 20.86, FID↓ = 4.37
						AFHQv2 CATS	FID↓ = 2.71
2023	Jo et al. [56]	✓	×	×	CG-NeRF, which can generate multi-view images reflecting extra input conditions such as images or texts	CelebA-HQ	FID↓ = 7.01
						Cats	FID↓ = 13.86
						CUB-200	FID↓ = 26.53
2023	Li et al. [57]	✓	×	×	3D-aware multi-class image-to-image translation with NeRFs	CelebA-HQ	FID↓ = 22.30
						AFHQ	FID↓ = 15.30
2023	Gao et al. [58]	✓	×	×	Sketch-based facial generation and editing in NeRFs	100 hand-drawn sketches	FID↓ = 72.63
2023	Shen et al. [60]	✓	×	×	Toward lifelike talking head animation via spatially-adaptive dual-driven NeRFs	High-resolution talking videos from YouTube (Test set A)	PSNR↑ = 34.40, SSIM↑ = 0.965, LPIPS↓ = 0.058
2023	Li et al. [61]	✓	×	×	Deformable NeRF model capable of one-shot learning and subject agnosticism	VoxCeleb1 dataset	PSNR↑ = 21.90, SSIM↑ = 0.862, LPIPS↓ = 0.084
2023	Li et al. [62]	✓	×	×	A conditional NeRF for talking portraits using tri-plane hashing, region attention, and adaptive pose encoding for efficiency and realism	Four high-definition speaking video clips from publicly-released video sets	Lip-sync accuracy = 4.14, Image quality = 4.08, Video realness = 3.86
2023	Hu et al. [66]	×	✓	×	A neural rendering pipeline combining volumetric and textural rendering to synthesize high-quality, pose-driven 3D human avatars	Self-collected dataset (Unseen poses from R7 – P-UV)	PSNR↑ = 27.92, SSIM↑ = 0.859, LPIPS↓ = 0.087

Table 3 (continued)

Year	Author(s)	F	B	H	Key concept	Dataset	Performance
2023	Yu et al. [67]	×	✓	×	High-fidelity, view-consistent avatar animation from monocular videos, leveraging a bidirectional deformation field and keyframe-guided correspondence	ZJU-MoCap dataset	NVS: PSNR↑ = 30.26, SSIM↑ = 0.969, LPIPS↓ = 30.920 NPS: PSNR↑ = 30.05, SSIM↑ = 0.968, LPIPS↓ = 31.510
2023	Chen et al. [68]	×	✓	×	A geometry-guided attention mechanism aimed at synthesizing detailed novel view images	THuman2.0 dataset	PSNR↑ = 28.88, SSIM↑ = 0.951, LPIPS↓ = 0.033
2023	Zhu et al. [69]	×	✓	×	CAT-NeRF, transformer-based dynamic body modeling with temporal appearance constancy	H36M	NVS: PSNR↑ = 24.57, SSIM↑ = 0.900 NPS: PSNR↑ = 23.68, SSIM↑ = 0.889
2023	Peng et al. [70]	×	✓	×	Fast NeRF training with optimizable intrinsic coordinates for human performance rendering	ZJU-MoCap Custom data	PSNR↑ = 25.86, SSIM↑ = 0.889 PSNR↑ = 26.21, SSIM↑ = 0.881
2023	Zhu et al. [72]	×	✓	×	A large-scale multiview and multipose dataset for 3D clothed human avatar reconstruction	MVP-HUMAN	NORMAL = 0.021, P2S (CM) = 1.341, CHAMFER = 1.719
2023	Wang et al. [73]	×	✓	×	Double-layer NeRF approach for human performance capture that separately tracks body and clothing motion	S4 of DeepCap	PSNR↑ = 23.06, SSIM↑ = 90.280
2023	Guo et al. [74]	×	×		Reconstruct accurate appearance and geometry with NeRF for interacting hands	InterHand2.6 M dataset (4 view)	PSNR↑ = 29.03, SSIM↑ = 0.955, LPIPS↓ = 0.084
2023	Qu et al. [75]	×	×	✓	Neural rendering and pose estimation for hand-object interaction with NVS	HandObject	PSNR↑ = 22.20, SSIM↑ = 93.71, LPIPS↓ = 0.059
2024	Liu et al. [76]	×	✓	×	Harnessing dynamic NeRF for large-scale motion- and view-change human-centric video editing	HOSNeRF and NeuMan	CLIPScore↑ = 31.31
2024	Huang et al. [77]	✓	×	×	Geometry-enhanced NVS with depth-aware training	FFHQ	G-NeRF 64^2: FID↓ = 26.04
2024	Feng and Singhal [78]	✓	×	×	3D face style transfer with NeRF and mesh rasterization for multi-view consistency	Collected video dataset of 8 subjects	Short Range: LPIPS $\times 10^{-2}$↓ = 0.290 Long Range: LPIPS $\times 10^{-2}$↓ = 0.380

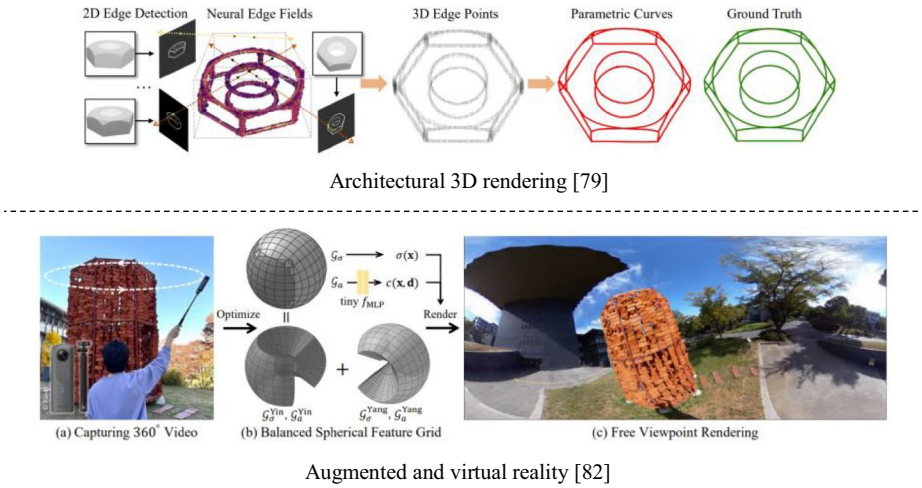


Fig. 7 An illustration showcasing the applications of radiance fields in rendering, including architectural 3D visualization and AR/VR

Augmented and virtual reality In response to the increasing demand for high-fidelity 3D content in VR and AR, Xu et al. [81] presented NeuralLift-360, a framework designed to convert 2D images into 3D objects with comprehensive 360° perspectives. This framework utilized depth-aware NeRF in conjunction with denoising diffusion models, guided by CLIP-based methodologies, to produce high-quality 3D content, outperforming existing baselines. However, it encountered limitations regarding resolution and the complexity of scenarios involving occlusion and multiple objects. Choi et al. [82] devised EgoNeRF, a robust methodology for the reconstruction of expansive real-world environments designed for VR applications, utilizing 360° video inputs. EgoNeRF successfully generated NeRFs by utilizing a balanced spherical feature grid that is synchronized with the rays of egocentric images. However, several challenges persist in real-world applications, such as photometric inconsistencies and inaccuracies in camera parameters. Song et al. [83] bring forward NeRFPlayer, an efficient framework for navigating 4D spatiotemporal spaces in VR. The deliberated framework emphasizes rapid reconstruction, compact modeling, and streamable rendering by implementing a decomposition of the spatiotemporal domain informed by temporal characteristics. By using neural fields for representation and regularization of distinct areas and a hybrid representation-based feature streaming scheme, NeRFPlayer achieved comparable or superior rendering quality and speed, reconstructing scenes in approximately 10 s per frame. However, it faces limitations in representing long-range repeated activities and operates under the assumption of uniform camera imaging configurations. Jiao et al. [84] advanced the NVS method by developing VH-NeRF, a photorealistic free-viewpoint video (FVV) captured using a sparse multi-camera array. This innovation enables real-time live streaming of expansive dynamic scenes for immersive VR and AR applications on mobile platforms. Rojas et al. [85] contributed Re-ReND, a framework designed for real-time rendering of pre-trained NeRF models on resource-constrained devices. The given method significantly improved rendering speeds, achieving over a 2.6-fold increase com-

Table 4 A comparative assessment of NeRF-driven approaches for rendering applications. *Abbreviations*: Arc 3D – Architectural 3D rendering and AR/VR – Augmented and virtual reality

Year	Author(s)	Arc 3D	AR/VR	Key concept	Dataset	Performance
2023	Ye et al. [79]	✓	×	A neural implicit field optimized with view-based rendering loss to reconstruct 3D feature curves from multi-view images without requiring 3D edge supervision	A-N	F-score↑ = 0.9044, IoU↑ = 0.8283
					A-N-L	F-score↑ = 0.9353, IoU↑ = 0.8766
2023	Erich et al. [80]	✓	×	Export objects' geometry by learning a probabilistic representation of radiance and density	Self-collected dataset (Tissues)	Mean error in CM = 0.14
2023	Xu et al. [81]	×	✓	A depth-aware NeRF using diffusion models to generate 360° 3D objects from a single image for XR	CO3Dv2	CLIP distance↓ = 0.4498
2023	Choi et al. [82]	×	✓	A spherical grid-based NeRF approach for reconstructing large-scale VR environments from short 360° video captures	OmniBlender Indoor (100k)	PSNR↑ = 33.11, SSIM↑ = 0.902, LPIPS↓ = 0.142
					Ricoh360 (100k)	PSNR↑ = 25.25, SSIM↑ = 0.763, LPIPS↓ = 0.286
2023	Song et al. [83]	×	✓	4D NeRF framework that decomposes spatiotemporal spaces for efficient VR scene	Plenoptic video	PSNR↑ = 30.29
2023	Jiao et al. [84]	×	✓	A photorealistic FVV framework for real-time VR/AR streaming using sparse multi-camera input and NeRFs	NA	NA
2023	Rojas et al. [85]	×	✓	A real-time NeRF rendering method for resource-constrained devices, leveraging mesh-based density extraction and matrix-factorized light fields for efficient processing	Synthetic 360°	Samsung: PSNR↑ = 29.00 Desktop: PSNR↑ = 30.10
					Unbounded 360°	Samsung: PSNR↑ = 17.90 Desktop: PSNR↑ = 18.00

pared to similar techniques, all while preserving high-quality output, which makes it applicable across a variety of platforms, including mobile phones and VR headsets.

Summary and trend analysis NeRF-based rendering techniques have significantly advanced photorealistic content generation in architectural visualization, remote surveying, and immersive AR/VR applications, as detailed in Table 4. Recent methods emphasize efficiency, real-time performance, and adaptability to dynamic or sparse inputs, enabling interactive experiences even on mobile platforms. However, challenges remain in handling complex textures, occlusions, and large-scale dynamic environments, as well as achieving high-resolution outputs with minimal latency. Future directions include improving temporal coherence, integrating semantic understanding, and optimizing models for hardware-constrained devices, paving the way for scalable, interactive, and intelligent 3D content generation across diverse real-world settings.

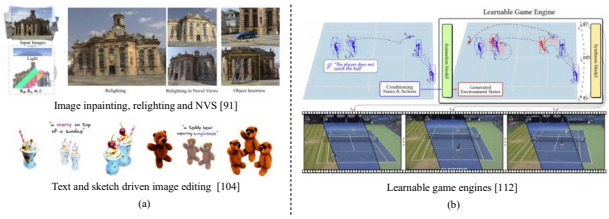


Fig. 8 Varied applications of radiance fields in interactive media development, including (**a**) multimedia editing and (**b**) video game design

3.1.2 Interactive media development

Interactive media development, including tasks such as image and video editing, video game design, and animation, has seen significant advancements. A notable breakthrough in this field has been the application of radiance field-based mechanisms, facilitating the generation of highly immersive and lifelike experiences, thereby expanding the possibilities in interactive media development. Figure 8 depicts the latest advancements in NeRF applications, specifically enhancing interactive media development.

a Multimedia editing
A major breakthrough in multimedia editing lies in the domains of style transfer between images, image inpainting, relighting, text-guided content editing, and etc. The following section explores these advancements in detail.

Style transfer In the domain of 3D style transfer, Nguyen-Phuoc et al. [86] presented a method for stylizing 3D scenes using neural implicit representations. The method introduced a training approach to address graphics processing unit (GPU) memory constraints, optimizing NeRF and stylization alternately to support higher-resolution outputs and advanced style transfer techniques. Meanwhile, Li et al. [87] expounded a ClimateNeRF, integrating physical simulations with NeRF models to render realistic weather effects, enhancing the visual realism of climate change simulations. However, the technique still faced challenges related to achieving real-time performance due to limitations in NeRF modeling. Zhang et al. [88] deployed LipRF, a framework for photorealistic 3D scene stylization that maintains both consistency and photorealism. The innovation includes preserving volume rendering variance, using Lipschitz MLP for control, and applying adaptive regularization and gradient aggregation. While versatile, its performance is heavily contingent upon the quality of the pre-trained radiance field, which may require further enhancements. Additionally, Huang et al. [89] deliberated NeRF-Texture, focused on capturing, modeling, synthesizing, and rendering textures from real-world scenes with intricate geometric and aesthetic characteristics. The study identified limitations associated with basic shape prerequisites for texture capture and potential semantic distortions during synthesis, recommending enhancements through the integration of advanced generative techniques.

Image inpainting Mirzaei et al. [90] bring forward a 3D inpainting technique tailored for editing NeRF scenes, focusing on the challenge of removing undesired objects while preserving visual realism and geometric integrity. The method utilized 3D segmentation alongside perceptual optimization to transfer insights from pre-trained 2D image inpaint-

ers into the 3D domain, delivering notable results in 3D inpainting and offering a valuable benchmark dataset. Wang et al. [91] proposed an inverse rendering framework tailored for extensive urban settings, enabling the concurrent reconstruction of scene geometry, spatially varying material properties, and high dynamic range lighting using RGB images with optional depth input. The stated approach combines a neural field with an explicit mesh structure, facilitating realistic relighting and the insertion of virtual objects complete with ray-traced shadows. While demonstrating considerable potential for applications in scene reconstruction and editing, this method faces challenges stemming from its dependence on manually crafted priors and its emphasis on static environments. Weder et al. [92] deployed a framework aimed at object removal from NeRFs generated from RGB-D sequences. The method employs user-supplied masks and a confidence-based view selection process to maintain 3D consistency. The proposed NeRF inpainting method leveraged single image inpainting and presented a new dataset for object removal and inpainting, though it acknowledged limitations related to mask size and challenges such as blurring and handling shadows or reflections. Tertikas et al. [93] put forward PartNeRF, a part-aware generative model for editable 3D shape synthesis, offering localized shape control without the need for explicit 3D supervision. Although PartNeRF made significant strides in producing editable 3D models with textures, it did not delve into more complex texture representations or extend its application to dynamic objects. Lazova et al. [94] unveiled Control-NeRF, a system that facilitates flexible manipulation of 3D-aware image content, generating high-quality NVS from posed input images. Control-NeRF utilized scene-specific 3D feature volumes alongside a general NeRF rendering network, enabling users to modify scenes by editing these volumes—such as mixing scenes or altering objects—while maintaining photorealistic output. Wang et al. [95] propounded Seal-3D, an editing method and system for NeRF allowing pixel-level editing with immediate previews, providing flexibility and speed for model editing. The proposed training strategy and editing instructions' proxy function enhanced user-friendly editing but possess limitations in handling complex lighting effects and addressing reconstruction failures within the original NeRF network. This could potentially be resolved through additional research, like introducing intrinsic decomposition. Wang et al. [96] developed a NeRF framework to model real-world scenes efficiently, allowing object editing and NVS in one process by separating 3D representations of objects and the background. This significantly improved rendering and editing quality, although there is a limitation in handling unclear background regions obstructed by objects. Kuang et al. [97] suggested PaletteNeRF, a technique that enhances NeRF's efficiency in photorealistic appearance editing through a palette-based color decomposition method. The proposed method enables intuitive modifications of 3D scene appearances by adjusting color palettes, showing improved performance across various editing tasks and making palette-based editing more practical in real-world scenarios. Jambon et al. [98] contributed NeRFshop, a tool designed for interactive object selection and manipulation within NeRFs, addressing predefined limitations in scene editing. NeRFshop features a scribble-based interface for object selection, semi-automatic cage creation, and interactive volumetric transformations, complemented by a membrane-based correction mechanism to mitigate artifacts, facilitating more adaptable scene adjustments. Additionally, Zhang et al. [99] proposed a point cloud-based method called Nerflets, which utilizes local NeRFs for efficient 3D scene representation from images. By optimizing Nerflets with both photometric and panoptic image supervision, this approach allows for scene decomposition, facilitating NVS, 3D segmenta-

tion, and interactive scene editing. However, it struggles with dynamic content modeling and complex media effects that span across semantic boundaries. Zheng et al. [100] deliberated an editable NeRF approach that supports dynamic scene editing and topology changes using a single-camera image sequence. The proposed method leverages key-point-driven radiance fields with a weighted strategy, offering intuitive multi-dimensional editing capabilities. Despite its effectiveness, it faces challenges with scene complexity, transparency, and key-point placement accuracy.

Scene relighting In the area of indoor and outdoor scene relighting, Yang et al. [101] introduced an approach to outdoor scene relighting, combining NeRFs and CNNs to estimate interpretable lighting parameters. The given method demonstrated improvements in intrinsic image decomposition and relighting, yet it acknowledged limitations in accurately handling cast shadows, which required user interaction or advanced graphics techniques for resolution. Whereas, Levy et al. [102] discussed a rendering model for NeRFs designed to account for the effects of scattering media, including underwater and foggy scenes. The method leveraged the SeaThru image formation model, allowing the separation of object appearance from medium effects, offering improved scene reconstruction, depth estimation, and the potential for simulating environments under various conditions. The research provided a significant extension to NeRFs, enabling the realistic rendering of scenes in scattering media, and while it had some limitations that needed to be addressed, such as transient effects and further diverse scenarios and medium parameters. Wu et al. [103] presented DE-NeRF, a framework that decouples view-independent and view-dependent appearances within NeRF, allowing for flexible scene manipulation and relighting. Despite these advancements, DE-NeRF struggles with handling thin structures, shadows, and inter-reflections.

Text guided editing Recent advancements have further enhanced 3D synthesis from text through direct NeRF optimization. Mikaeili et al. [104] unveiled SKED, a method for editing 3D shapes represented by NeRFs using only two guiding sketches from distinct viewpoints. Despite its innovative approach, SKED, like former methodologies, encounters difficulties with the multiface issue and relies on the publicly available stable diffusion model, which at times yields lower-quality 3D outputs compared to commercial diffusion models. Bao et al. [105] developed a semantic-driven editing method for NeRFs, enabling users to alter NeRFs using a single image to produce high-quality, multi-view consistent outputs. However, this method could not accommodate changes in object topology and required user-edited images with semantically rich content, which limited its effectiveness for images with random or less meaningful content. Song et al. [106] deliberated Blending-NeRF, a model designed for localized, text-guided editing of 3D objects. Blending-NeRF integrated pretrained and editable NeRF networks with novel blending operations, allowing for accurate and natural modifications of specific 3D object regions while maintaining their original appearance, and demonstrated superior results in text-guided 3D editing tasks. Wang et al. [107] proposed NeRF-Art, a text-driven NeRF stylization technique that modifies the style of a pre-trained NeRF model using simple text prompts. The suggested approach employs a global-local contrastive learning strategy, along with a directional constraint to regulate style transformation in terms of trajectory and intensity, allowing for variations in geometry and appearance without necessitating mesh guidance. Metzer et al. [108] focused on shape-guided

generation of 3D shapes and textures, particularly using LDMs in a compact latent space. The method adapt score distillation to LDMs, creating Latent-NeRF, which combines text and shape guidance for enhanced 3D object generation. The approach demonstrates the potential for rendering in the latent space of neural models, promoting further exploration of effective text-guided shape generation and rendering techniques.

Object detection and segmentation Xu et al. [109] put forward NeRF-Det, a method for 3D object detection in indoor environments derived from posed RGB images, utilizing multi-view geometry constraints through a shared geometry MLP. Its joint-training methodology facilitated generalization to unseen scenes for tasks such as object detection, view synthesis, and depth estimation without requiring per-scene optimization. Xu et al. [110] deployed MonoNeRD, a monocular 3D object detection framework that integrates NeRF with 3D detection, optimizing intermediate 3D feature representations through implicit NeRF-like methods. MonoNeRD set a benchmark for monocular 3D detection, though it encountered challenges in predicting 3D occupancy in specific environments like the sky. Hu et al. [111] offered NeRF-RPN, an innovative framework tailored for direct detection of 3D bounding boxes in NeRF environments, thereby obviating the necessity to render multiple viewpoints. The method demonstrate versatility with various architectures and loss functions, offering a benchmark dataset for future 3D object detection research. Further, NeRF-RPN's efficiency is evidenced by its quick training time and high recall rates on the 3D-FRONT NeRF dataset.

b **Video game design**
Menapace et al. [112] unveiled a framework for training learnable game engines (LGEs) solely from monocular annotated videos, capable of maintaining and rendering scenes, objects, and agents while modeling the logic and physics of a game. The suggested framework also integrates a unique synthesis model and animation system, allowing for high-quality, high-resolution video generation and precise control over complex game logic and interactions, offering applications beyond existing neural video game simulators. Additionally, the researchers presented large-scale annotated Tennis and Minecraft datasets, demonstrating the potential of learning game engines for real-world scenes and intricate applications. Sin et al. [113] contributed NeRFahedron, a neural rendering pipeline designed to facilitate interactive neural rendering for animatable 3D content. By localizing NeRF fields in tetrahedrons and implementing optimized techniques, the method showcased real-time performance, addressing the challenge of interactive applications, while acknowledging the necessity for further research on complex scenes and semi-transparent surfaces.

Summary and trend analysis NeRF have emerged as a transformative technology in interactive media development, enabling photorealistic 3D content creation, editing, and rendering from sparse inputs. Recent advancements have expanded NeRF's capabilities beyond static scene reconstruction to dynamic, editable, and stylized representations, supporting applications in multimedia editing, video game design, and virtual world building. The analysis presented in Table 5 reveals that most interactive media development techniques predominantly focus on multimedia editing. These methodologies encompass a range of tasks,

Table 5 An analysis of radiance field-based mechanisms for the domain of interactive media development. *Abbreviations*: ME – Multimedia editing and VGD – Video game design

Year	Author(s)	ME	VGD	Key concept	Dataset	Performance
2022	Nguyen-Phuoc et al. [86]	✓	×	A stylized NeRF training method that alternates NeRF and stylization optimization to achieve high-quality, cross-view consistent NVS	Tanks and Temples dataset (Train)	Short-range consistency: Consistency score↓ = 0.008
2023	Li et al. [87]	✓	×	A NeRF-editing method that fuses physical simulations with scene models to realistically visualize climate change effects	Tanks and Temples, NeRF360 and KITTI-360	NA
2023	Zhang et al. [88]	✓	×	Photorealistic 3D scene stylization by leveraging Lipschitz mapping for cross-view consistency	LLFF dataset	Long temporary consistency↑ (WCT2 w/Lip) = 25.90
2023	Mirzaei et al. [90]	✓	×	Leveraging 2D image inpainters and perceptual optimization for removing objects from NeRF scenes while ensuring multi-view consistency and geometric validity	Real-world scenes from LLFF and NeRF360	Inpainting method, using the outputs of multiview segmentation model: LPIPS↓ = 0.466, FID↓ = 140.56
2023	Wang et al. [91]	✓	×	Hybrid neural field and mesh-based approach for high dynamic range lighting for realistic relighting and virtual object insertion	NeRF-OSR dataset	Scene relighting (Site 1): PSNR↑ = 21.53, MSE↓ = 0.007
2023	Weder et al. [92]	✓	×	User-guided method for removing objects from NeRFs using 2D image inpainting and confidence-based view selection to ensure 3D consistency	Synthetic objects	PSNR↑ = 29.43, SSIM↑ = 0.916, LPIPS↓ = 0.070
2023	Tertikas et al. [93]	✓	×	Part-aware generative model using locally defined NeRFs with affine transformations, enabling 3D shape editing without explicit supervision	Airplane Table	MMD-CD↓ = 1.37, COV-CD↑ = 37.90 MMD-CD↓ = 4.48, COV-CD↑ = 40.60
2023	Lazova et al. [94]	✓	×	Control-NeRF combines scene-specific 3D feature volumes with a fixed NeRF rendering network, enabling flexible scene editing	LLFF dataset	PSNR↑ = 25.63, SSIM↑ = 0.853, LPIPS↓ = 0.181
2023	Wang et al. [95]	✓	×	Seal-3D enables fast, pixel-level NeRF editing with instant previews using a proxy function and two-stage training	Blender synthetic dataset	PSNR↑ = 35.08
2023	Wang et al. [96]	✓	×	A unified NeRF framework for joint scene decomposition and composition enables high-quality NVS and object-level editing	ToyDesk ScanNet	PSNR↑ = 23.14, SSIM↑ = 0.789, LPIPS↓ = 0.469 PSNR↑ = 26.13, SSIM↑ = 0.824, LPIPS↓ = 0.395

Table 5 (continued)

Year	Author(s)	ME	VGD	Key concept	Dataset	Performance
2023	Kuang et al. [97]	✓	×	A rotation-invariant point-based representation for fine-grained editing and cross-scene compositing	NeRF synthetic dataset and ScanNet dataset	PSNR↑ = 31.81, SSIM↑ = 0.956
2023	Zhang et al. [99]	✓	×	Nerflets are local NeRFs enabling efficient 3D scene representation, segmentation, and interactive editing	ScanNet	PSNR↑ = 29.12
2023	Zheng et al. [100]	✓	×	An editable NeRF for dynamic scenes, enabling intuitive user edits and topological changes via key point manipulation	Synthetic dataset	Editing: PSNR↑ = 40.01, MS-SSIM↑ = 0.995, LPIPS↓ = 0.082
2023	Yang et al. [101]	✓	×	Combines NeRF and CNNs for interpretable, editable lighting decomposition from single images	Phototourism and Mega-Depth dataset	PSNR↑ = 24.23, SSIM↑ = 0.855, LPIPS↓ = 0.135
2023	Levy et al. [102]	✓	×	Extends NeRF to underwater and foggy scenes using SeaThru, enabling realistic novel views and medium removal	Red sea	PSNR↑ = 21.83, SSIM↑ = 0.770, LPIPS↓ = 0.250
2023	Wu et al. [103]	✓	×	Decoupled NeRFs for view-consistent appearance editing and high-frequency environmental relighting	NeRF synthetic	PSNR↑ = 29.18, SSIM↑ = 0.959, LPIPS↓ = 0.035
2023	Mikaeili et al. [104]	✓	×	A text and sketch-guided NeRF editing method enabling intuitive 3D shape manipulation with minimal user input	Require only text and sketch inputs from the user	Sketch alignment score: IoS↑ = 0.8220 Semantic alignment score: CLIP-similarity↑ = 0.2739
2023	Bao et al. [105]	✓	×	Semantic-driven NeRF editing enabling high-fidelity, multi-view consistent 3D edits from a single image using prior-guided fields and geometric constraints	Real-world car dataset, PhotoShape, NeRF 360°, NeRF Synthetic, DTU, BlenderSwap	89.5%/83.3%
2023	Song et al. [106]	✓	×	Blending-NeRF leverages a dual-NeRF architecture and CLIP-based guidance for localized text-driven 3D object editing	Realistic synthetic 360◦ dataset	MPCLIP↑ = 0.121
2023	Wang et al. [107]	✓	×	NeRF-Art enables text-driven NeRF stylization by altering both geometry and appearance with contrastive learning and directional constraints	Self-collected dataset	CLIP-similarity↑ = 0.7356

Table 5 (continued)

Year	Author(s)	ME	VGD	Key concept	Dataset	Performance
2023	Xu et al. [109]	✓	×	NeRF for indoor 3D object detection, integrating geometry priors and a shared MLP for efficient, generalizable detection without per-scene optimization	ScanNet	NeRF-Det-R101-2x*: mAP@0.25 = 53.30 (+4.30)
					ARKITScenes dataset	NeRF-Det-R50: mAP@0.25 = 26.70 (+3.10)
2023	Xu et al. [110]	✓	×	Volume rendering for monocular 3D detection, using signed distance functions (SDF)-based NeRF representations to infer dense 3D geometry	KITTI-3D	AP_{BEV} (Hard) = 20.97, AP_{3D} (Hard) = 15.63
					Waymo	Level 1: 3D mAP/mAPH (IoU↑ = 0.7) (50 m - ∞) = 0.72/0.71
2023	Hu et al. [111]	✓	×	NeRF-RPN detects 3D objects directly from NeRF using multi-scale volumetric features, enabling class-agnostic detection without rendering	ScanNet	Recall (anchor-free) = 18.40
2023	Menapace et al. [112]	×	✓	Compositional NeRF and a learned animation model for AI-driven gameplay and interactive rendering from monocular videos	Self-collected Tennis dataset	LPIPS↓ = 0.152, FID↓ = 12.80
					Self-collected Minecraft dataset	LPIPS↓ = 0.008, FID↓ = 2.81
2023	Sin et al. [113]	×	✓	Enhancing real-time interactivity by localizing NeRF fields, reducing computational overhead, and enabling animatable 3D content for seamless integration into gaming environments	Synthetic dataset	Sonic-Walk: PSNR↑ = 40.43, LPIPS↓ = 0.039, MSE↓ = 5.91
2024	Otonari et al. [114]	✓	×	Enhances NeRF for dynamic urban scenes by removing moving objects using entity-wise statistics and segmentation	MovieMap	Entity-NeRF with Nerfacto: PSNR↑ = 24.93, SSIM↑ = 0.850, LPIPS↓ = 0.120
2024	Chen et al. [115]	✓	×	Removes transient distractors in NeRF using heuristics-guided segmentation, blending SfM-based and color residual heuristics with advanced segmentation for cleaner scene reconstruction	Kubric dataset	mIoU↑ = 0.944, F1↑ = 0.934
2024	Zhang et al. [116]	✓	×	Large-scale segmentation models and hierarchical embeddings to enable accurate 3D object segmentation from open-vocabulary queries	Scene: Desktop	Pixel-wise area under the precision recall curve (AUPRC) = 88.90 $_{\uparrow 10.1}$, FPR = 3.32 $_{\uparrow 6.8}$
2024	Fischer et al. [117]	✓	×	NeRF analogies transfers appearance from a source NeRF onto a target 3D geometry while preserving semantic meaning	Mip-NeRF 360 and Tanks and Temples	CLIP↑ = 0.992

Table 5 (continued)

Year	Author(s)	ME	VGD	Key concept	Dataset	Performance
2024	Chen et al. [118]	✓	×	Enhances NeRF inpainting by leveraging diffusion priors for view-consistent appearance and geometry completion	Self-collected Real-S Self-collected Real-L	PSNR↑ = 17.66, LPIPS↓ = 0.507 PSNR↑ = 25.69, LPIPS↓ = 0.181

Autonomous navigation [120] Robotics [122]

Fig. 9 Deployment of radiance fields in automated systems, including autonomous navigation and robotics

such as style transfer, image morphing, object detection, and segmentation, as illustrated in the accompanying comparison. Key trends include the integration of generative models for text-guided editing, acceleration techniques for real-time performance, and hybrid representations that combine NeRFs with meshes or point clouds for better control and simulation. The rise of multimodal interfaces, real-time rendering engines, and personalized 3D avatars signals a shift toward democratized content creation and immersive user experiences. As research continues to improve editability, efficiency, and platform compatibility, NeRF is poised to become a foundational technology for interactive applications in the metaverse and beyond.

3.1.3 Automated systems

Autonomous systems, such as self-driving cars and robotics, rely on perceiving and understanding the environment to make informed decisions. NeRF technology, which reconstructs detailed 3D scenes from 2D images, holds potential for enhancing autonomous systems. By enabling these systems to create highly detailed and accurate 3D representations of their surroundings, NeRF could improve its perception abilities. This technology can aid in better understanding the environment, predicting and simulating scenarios, and allowing for more informed decision-making in navigation, object interaction, and obstacle avoidance. Figure 9 showcases applications of radiance fields in the domain of automated systems. Moreover, NeRF's capacity for generating novel views could assist in simulating potential scenarios, enabling autonomous systems to make better predictions and decisions in unfamiliar or dynamically changing environments.

a **Autonomous navigation**

In 2022, Huang et al. [119] proposed a sophisticated methodology that enhances NeRF for large-scale indoor environments by employing geometry-aware techniques. This advancement facilitates the generation of high-quality 360° imagery suitable for real-time VR navigation, demonstrating superior photorealistic rendering capabilities compared to traditional 3D reconstruction methods. The approach indicates significant potential for delivering immersive VR experiences in expansive indoor settings while minimizing resource consumption. Deng et al. [120] put forward NeRF-LOAM, a pioneering neural implicit approach for odometry and mapping in extensive environments that leverages LiDAR data. The suggested approach achieved comparable results in both odometry and mapping tasks while exhibiting robust generalization across various environments, though it still requires optimization for real-time applications.

b **Robotics**

In the fields of robotics, Ze et al. [121] introduced GNFactor, a visual behaviour cloning agent for multi-task robotic manipulation, which utilizes a generalizable neural feature field in conjunction with the perceiver transformer architecture. Although GNFactor exhibits robust task generalization, its reliance on multiple fixed cameras poses scalability issues. In contrast, Zhou et al. [122] propounded SPARTN, a data augmentation technique designed to enhance robotic policies by employing eye-in-hand camera systems. SPARTN utilizes NeRFs to enhance robotic grasping success rates in both simulations and real-world applica-

Table 6 A brief comparison among radiance field-based methods in automated systems. *Abbreviations*: AN – Autonomous navigation and R – Robotics

Year	Author(s)	AN	*R*	Key concept	Dataset	Performance
2022	Huang et al. [119]	✓	×	NeRF for large-scale scenes with geometry-aware optimization, improved positional encoding, and adaptive refinement for roaming in large scale indoor scenes	Real Scenes	PSNR↑ = 24.67, SSIM↑ = 0.755, LPIPS↓ = 0.209
					Synthesis	PSNR↑ = 36.28, SSIM↑ = 0.955, LPIPS↓ = 0.046
2023	Deng et al. [120]	✓	×	NeRF-LOAM jointly optimizes LiDAR odometry and mapping using a neural signed distance function, reducing drift and enhancing large-scale reconstruction	MaiCity	RMSE↓ (Mai01) = 0.13
					Newer College	RMSE↓ = 0.15
					KITTI	RMSE↓ = 4.26
2023	Wu et al. [123]	✓	×	NeRF-based autonomous driving simulator with instance-aware modeling, modular design, and photorealism	KITTI-75%	NVS: PSNR↑ = 24.23, SSIM↑ = 0.845, LPIPS↓ = 0.160
2023	Ze et al. [121]	×	✓	A generalizable neural feature field for multi-task robotic manipulation, integrating 3D voxel representations with vision-language semantics	Self-created scene for real robot	Evaluate 10 episodes for each task and report the resulting success rate (%) = 43.30
2023	Zhou et al. [122]	×	✓	A NeRF-based offline augmentation method that injects corrective noise into visual demonstrations, enhancing RGB-only robotic manipulation without additional expert supervision	Robot setup	Unseen test object: BC SR (%) = 37, SPARTN SR (%) = 58

tions, though it faces challenges related to computational costs and platform-specific noise adjustments.

Summary and trend analysis NeRF-based methods are increasingly being integrated into automated systems, enhancing environmental perception and decision-making in both autonomous navigation and robotics. Table 6 presents an analysis of the latest advancements in automated system development. In navigation, NeRF variants like those by Huang et al. [119] and Deng et al. [120] demonstrate the ability to reconstruct large-scale indoor and outdoor environments with high fidelity, enabling realistic VR navigation and accurate odometry through LiDAR-NeRF fusion. In robotics, approaches such as GNFactor [121] and SPARTN [122] leverage NeRFs for multi-task manipulation and policy augmentation, improving generalization and task success rates despite challenges in scalability and computational cost. The overarching trend reveals a shift toward hybrid systems that combine NeRFs with geometric priors, LiDAR, and transformer architectures, aiming to balance photorealism, real-time performance, and generalization. Future directions suggest a focus on optimization for real-time deployment, reduced resource requirements, and improved adaptability across varying physical platforms and dynamic scenes.

3.1.4 Visual computing

Visual computing, including medical imaging and signal modeling, leverages NeRF to convert 2D data into detailed 3D representations. Figure 10 showcases advancements in NeRF applications within the field of visual computing. This technology enhances accuracy in medical image reconstruction and offers improved 3D signal modeling capabilities for applications like audio synthesis, paving the way for advancements in diagnostics and signal processing. Additionally, Table 7 provides a comparison analysis of SOTA NeRF-based visual computing mechanisms.

a Medical imaging

In 2023, Petkov [124] trained NeRF on pre-rendered photorealistic imagery to facilitate a more efficient representation of complex medical datasets. The stated research also introduced interactive applications that enable the synthesis of novel perspectives directly from neural networks, showcasing a practical implementation for high-resolution computed tomography (CT) volume data using differentiable rendering techniques for view selec-

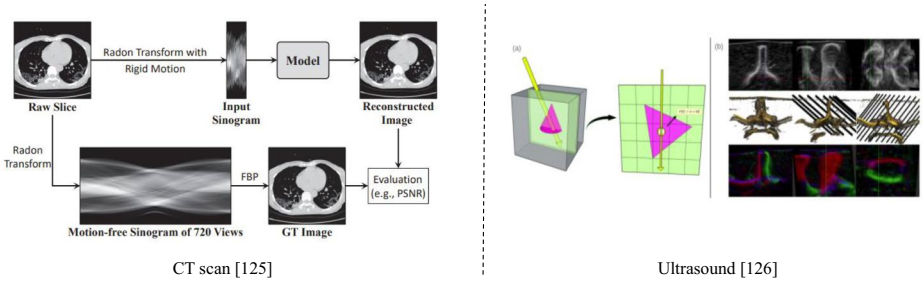


Fig. 10 A representation of various applications of radiance fields in medical imaging

Table 7 An analysis of SOTA NeRF based mechanisms in visual computing. *Abbreviations*: MI – Medical imaging and SM – Signal modeling

Year	Author(s)	MI	SM	Key concept	Dataset	Performance
2023	Petkov [124]	✓	×	Uses NeRFs to compress high-resolution CT data and enable interactive NVS via differentiable rendering	Ex-vivo 60GB volume dataset	Reduces the pixel RMSE↓ by 3% and the RMS of the perceptual hash difference by 58% compared to the path tracing
2023	Wu et al. [125]	✓	×	Enhances INR-based sparse-view CT by jointly optimizing transformation matrices and CT images to correct rigid motion artifacts	COVID-19 dataset	PSNR↑ = 23.36, SSIM↑ = 0.800, LPIPS↓ = 0.036
2023	Shi et al. [127]	×	✓	Uses a low-rank constrained NeRF with ADMM optimization and tensor train decomposition to achieve efficient light field compression with superior rate-distortion performance	HCI dataset	BD-PSNR = 1.63
2023	Liang et al. [128]	×	✓	Integrates NeRF with an acoustic-aware module to generate realistic spatial audio-visual experiences	Real-world audio-visual scene dataset	MAG = 1.504, ENV = 0.145
2024	Wysocki et al. [126]	✓	×	NeRFs for ultrasound imaging	Liver synthetic	With rendering (Median): SSIM↑ (Tilted) = 0.470, SSIM↑ (Perpendicular) = 0.490
					Spine phantom	With rendering (Median): SSIM↑ (Tilted) = 0.540, SSIM↑ (Perpendicular) = 0.540

tion and recognition. Wu et al. [125] introduced a self-calibrating neural field for sparse-view computed tomography (SVCT) reconstruction, significantly outperforming existing NeRF-based SVCT methods in robust and high-quality CT image reconstruction from motion-corrupted measurements. However, it currently supports 2D parallel X-ray beam CT and cannot handle complex non-rigid motion. Additionally, Wysocki et al. [126] presented Ultra-NeRF, an implicit neural representation designed for ultrasound imaging that synthesizes accurate B-mode images while accounting for the anisotropic characteristics of ultrasound, thereby improving ultrasound image synthesis and facilitating its integration into medical applications.

b **Signal modeling**

Shi et al. [127] bring forward a novel method for compressing light fields by employing a low-rank-constrained neural scene representation. This approach diverges from conventional techniques by first training an MLP-based NeRF using the provided input views. The research also focuses on optimizing the MLP parameters while adhering to a low-rank constraint, applying the alternating direction method of multipliers to achieve an efficient compression of the NeRF scene representation. The results demonstrated superior rate-distortion performance and free-viewpoint reconstruction capabilities compared to reference methods,

showcasing promise for achieving high compression ratios while maintaining view coherence and reconstruction quality. Liang et al. [128] tackled the challenge of generating real-world audio-visual scenes with spatial audio using NeRF. The model incorporated acoustic awareness into NeRF, enabling the generation of binaural audio for novel camera poses. Further comparative results showcased its effectiveness on both FAIR-PLAY and real-world indoor scenes. Future explorations involve delving into sound propagation, managing dynamic sound sources, and integrating the model with audio-visual simulation platforms to enhance robustness and generalization.

Summary and trend analysis While NeRF shows significant promise in visual computing, current approaches face notable limitations that hinder broader deployment. In medical imaging, despite improvements in reconstruction quality and interactivity, existing models often struggle with non-rigid motion and are limited to specific modalities like 2D parallel-beam CT or controlled ultrasound settings. Similarly, in signal modeling, although low-rank constraints and acoustic-aware NeRFs demonstrate high compression efficiency and immersive audio-visual synthesis, challenges remain in scaling to real-time applications, handling dynamic audio sources, and ensuring robustness across diverse environments. These limitations point to key future directions: developing generalized frameworks that support multi-modal and dynamic inputs, enhancing real-time performance, and integrating physics-informed priors to improve accuracy and interpretability. Additionally, expanding datasets and benchmarks tailored for these applications will be critical to validate and drive progress in clinical and real-world multimedia environments.

3.1.5 Others

Recent advancements in implicit mapping and positioning demonstrated favourable outcomes by utilizing NeRF characteristics. NeRFs facilitated the representation of continuous volumetric density and RGB values within a neural network, aiding in reconstructing unknown scene geometries. Beyond the applications outlined in the preceding section, NeRF have demonstrated versatility across diverse fields, including aerospace imaging, data augmentation, underwater exploration and etc. Alongside, to address the issues associated with feature extraction and correlation in the 3D reconstruction of spatial targets obtained from radar image sequences, Bintsi et al. [129] utilized NeRFs to create a physically-consistent 3D representation of the sun, focusing on extreme ultraviolet emission. The capability to reconstruct non-ecliptic viewpoints from observations limited to the solar ecliptic. The SuNeRF model attained SOTA results in 3D representations of the sun, laying the groundwork for potential applications in satellite observations of the sun, comparison with other stars, and improvements in space-weather forecasting, with future scope to expand SuNeRFs to output physical quantities such as plasma density and temperature. Liu et al. [130] developed RaNeRF, a reconstruction method that leverages inverse synthetic aperture radar (ISAR) image sequences. The given optimized framework enables the synthesis of ISAR images from novel observational angles, supporting applications in data augmentation and target analysis. Lei et al. [131] unveiled SAR-NeRF, a model merging SAR imaging principles with neural networks, implicitly modeling SAR images within a 3D imaging space. Quantitative experiments demonstrated SAR-NeRF's effectiveness in multi-view representation and significantly improved SAR target classification under a few-shot learning setup.

Marí et al. [132] suggested EO-NeRF, a methodology for digital surface modeling and next-generation visualization systems employing multi-temporal remote sensing imagery. EO-NeRF's proficient shadow rendering and its ability to process raw satellite images position it as an advanced tool for earth observation initiatives. Gableman and Kak [133] built upon the progress of Shadow NeRF and Sat-NeRF to develop a NeRF capable of rendering season-specific features independently of viewing and solar angles while retaining shadow-rendering capabilities. The proposed method achieved this by introducing the time of the year as an additional input variable and refining the network with loss functions to discourage the use of seasonal features for shadow rendering, demonstrating its effectiveness on satellite imagery captured by the Maxar WorldView-3 satellite across eight areas of interest. Giusti et al. [134] introduced MaRF, a novel framework for synthesizing the martian environment using spacecraft camera images, focusing on planetary geology, navigation, and shape analysis. MaRF utilized NeRFs to optimize a continuous volumetric scene function from sparse images, offering a more efficient and adaptable approach, demonstrated through evaluations on datasets from curiosity rover, perseverance rover, and ingenuity helicopter, potentially advancing scientific exploration and collaborative space missions by providing better insights into remote, unexplored environments. Xie et al. [135] proposed RS-NeRFs, an approach for multi-view satellite photogrammetry, aimed at swiftly generating accurate elevation predictions and novel view images for earth observation. They accomplished this through efficient adaptive sampling methods, a lightweight network with HashEncoder, and image inpainting techniques to eliminate artifacts. This significantly reduced training times and improved performance in elevation estimates and NVS while maintaining a mean absolute error of about 2 m for altitude predictions.

Tong et al. [136] presented Drive-3DAug, an innovative 3D data augmentation technique for camera-based 3D perception in autonomous driving. This approach utilized NeRF to create 3D models of background and foreground objects, resulting in enhanced detection accuracy on Waymo and nuScenes datasets, while providing adaptable digital driving assets for the research community. Ling et al. [137] deliberated automatic data factory (ADF), a self-supervised optical flow training framework, that leverages NeRF to generate high-quality optical flow labels from monocular images, eliminating the need for manual annotation or specialized equipment. By reconstructing scenes and filtering erroneous labels using optical flow matching, radiance field confidence, and depth consistency, ADF significantly enhances real-world generalization. The proposed ADF58 dataset, comprising 58,800 images from 300+scenes, demonstrates superior zero-shot generalization, surpassing existing self-supervised and supervised methods. Chu et al. [138] bring forward a technique for accurately reconstructing dynamic fluid phenomena from sparse multi-view RGB video data, without necessitating prior knowledge of lighting or geometric parameters. This method employs a spatio-temporal neural representation, physics-informed learning, and a hybrid architecture to differentiate dynamic fluid scenes from static obstacles, marking a significant progression in fluid dynamics reconstruction applicable to real-world scenarios and suggesting potential uses in real-time fluid capture and complex fluid-obstacle interactions. Nguyen et al. [139] developed CNT-NeRF, a GAN specifically designed for the simultaneous decomposition and segmentation of depth layers in carbon nanotube (CNT) forests from scanning electron microscope imagery. CNT-NeRF converts a single two-dimensional image into a multi-layered 2.5D representation, substantially reducing image complexity while enhancing the segmentation of CNTs, even in the presence of occlusions and com-

plex three-dimensional structures. The network, trained on photorealistic synthetic datasets, demonstrated significant capabilities in depth layer decomposition, achieving a 7–15% improvement in CNT segmentation compared to traditional U-Net segmentation methods. This facilitates the automated and non-destructive characterization of the physical properties of CNT forests, contributing to advancements in material discovery.

3.2 NeRF architectural modification

Despite NeRF's efficiency in generating novel views of static and dynamic scenes, its initial release required significant memory and time. Recent advancements focus on improving energy efficiency, speed, reduced training data, and high-quality novel views. Table 8 provides a comprehensive analysis of SOTA NeRF architectural advancements, highlighting key improvements, optimizations, and their impact on performance and application domains.

3.2.1 Energy efficiency

In 2022, Zheng et al. [140] introduced a resistive random access memory (RRAM)-based accelerator for NeRF rendering, designed to enhance the performance and energy efficiency of photorealistic scene rendering. On the other hand, Rao et al. [141] developed an energy-efficient, precision-scalable MAC array for NeRF rendering that supports variable weight bit-widths and has energy efficiency of 14.54 TOPS/W for 4-bit and 4.83 TOPS/W for 8-bit operations. Also NeRPIM, an energy-efficient 3D neural rendering processing-in-memory (PIM) processor, is suggested by Jo et al. [142] for mobile devices. Where a PIM-based renderer reduces the per-sample rendering energy with 8 T SRAM PIM. Long et al. [143] proposed a method to enhance the precision-scalable calculation efficiency of NeRF rendering. For NeRF models, the technique first analyzes and validates the concept of scene-dependent quantization. On the basis of this, it suggests designs for precision-scalable computation units based on look-up table (LUT) processing elements (PE). The mechanism implements these designs and compares the corresponding area, power, speed, and energy efficiency to assess the performance of several precision-scalable computing units.

3.2.2 Enhanced processing speed

In 2023, Zhu et al. [144] put forward the Pyramid NeRF, which improved training and inference speed for NeRF by 15x and 805x, respectively, by employing organized frequency-guided optimization and progressive subdivision. Also, Rivas-Manzaneque et al. [145] propounded NeRFLight, a grid-based model designed for real-time view synthesis, capable of over 150 FPS. By dividing the density field into regions and reusing feature grids, NeRFLight accelerates rendering while ensuring high-quality output, making it ideal for bandwidth-limited applications and metaverse scenarios. A systematic distillation technique, progressive volume distillation (PVD) is offered by Fang et al. [146] that allows for conversions across various NeRF architectures. With PVD an MLP-based NeRF model can be distilled from a hash table-based Instant-NGP model at a 10×∼20× faster speed than being trained the original NeRF from scratch, while achieving a superior level of synthesis quality. Learnable grid-based blur kernels and a sharpness prior are utilized to optimize spa-

Table 8 A comparative analysis of SOTA nerf's architectural modifications. *Abbreviations*: EE – Energy efficiency, PS – Enhanced processing speed, MS – Minimized storage requirement, and HFNVS – High-fidelity novel view synthesis

Year	Author(s)	EE	PS	MS	HFNVS	Key concept	Performance
2022	Zheng et al. [140]	✓	×	×	×	RRAM-based accelerator designed to enhance the performance and energy efficiency of photorealistic scene rendering	Energy efficiency (×10−4FPS/W)=2225.76
2022	Rao et al. [141]	✓	×	×	×	Energy-efficient precision-scalable MAC array for NeRF rendering	Energy efficiency of 14.54 TOPS/W and 4.83 TOPS/W for 4-bit and 8-bit operations
2023	Jo et al. [142]	✓	×	×	×	Energy-efficient 3D neural rendering PIM processor for mobile devices	Energy efficiency of 4.2 mJ/frame
2023	Long et al. [143]	✓	×	×	×	Analyzes and validates the concept of scene-dependent quantization, then suggests designs for precision-scalable computation units based on LUT PE	Energy efficiency of 1.84 TOPS/W and 1.38 TOPS/W for 12-bit and 16-bit operations
2023	Zhu et al. [144]	×	✓	×	×	Improves training and inference speed by employing organized frequency-guided optimization and progressive subdivision	Training and inference speed at 15× and 805×
2023	Rivas-Manzanequet al. [145]	×	✓	×	×	Divides the density field into regions and reusing feature grids, to accelerate rendering while ensuring high-quality outputs	Real-time view synthesis, capable of over 150 FPS
2023	Fang et al. [146]	×	✓	×	×	A structured distillation technique that allows for conversions across various NeRF architectures	MLP-based NeRF model can be distilled from a hash table-based Instant-NGP model at a 10×∼20× faster speed than being trained the original NeRF from scratch
2023	Li et al. [151]	×	×	✓	×	Leverages the smooth and continuous nature of viewpoint changes, using feature map with reduced resolution and a compact 2D neural renderer	Memory in MB (INT8)=27.30
2023	Zhao et al. [152]	×	×	✓	×	A three-stage pipeline that reduces the storage demand for voxel radiance fields while preserving speed and synthesis quality by leveraging frequency domain transformation, pruning, and quantization	Params (MB) of 10, 4 and 2 for TinyNeRF (10 MB), (4 MB), and (2 MB) respectively
2023	Chen et al. [153]	×	×	✓	×	BOF-NeRF to address the challenge of estimating correct geometry and synthesizing novel views	Requires 5.1G video memory

Table 8 (continued)

Year	Author(s)	EE	PS	MS	HFNVS	Key concept	Performance
2023	Takeda et al. [154]	×	×	✓	×	Use a photometric bundle adjustment to train NeRF, allowing it to optimize both camera poses and radiance fields at the same time	The time required for pose estimation is successfully accelerated by about 100 times by reducing the input image to COLMAP by about 1/10
2023	Zhao et al. [155]	×	×	✓	×	Leverages near-memory processing to overcome memory-bound bottlenecks in NeRF training	Parameter/data sizes for iNGP's bottleneck steps: HT = 25 MB, MLP^* = 0.014 MB, $MLP_{_b}^*$ = 0.014 MB, and $HT_{_b}$ = 25 MB
2023	Rho et al. [156]	×	×	✓	×	Minimize the size of grid-based neural fields used in neural rendering by applying a wavelet transform and a trainable masking approach	2 MB memory budget by removing over 95% of grid parameters while maintaining visual quality
2023	Gai et al. [159]	×	×	×	✓	Address the issue of blurry renderings and aliasing artifacts that occurs in certain textures or edges produced by NeRF	Enhances the average PSNR↑, SSIM↑, and LPIPS↓ performance on Blender dataset for NeRF by 0.30, 0.020, and 0.023 and Instant-NGP's by 0.23, 0.001, and 0.001 respectively
2023	Bian et al. [160]	×	×	×	✓	Optimizes NeRF and camera poses simultaneously, allowing for improved performance in handling challenging camera trajectories	Mean PSNR↑, SSIM↑, and LPIPS↓ for NVS results on ScanNet and Tanks and Temples is 31.86, 0.830, and 0.380 and 26.34, 0.740, and 0.390 respectively
2023	Isaac-Medina et al. [161]	×	×	×	✓	Tackle issues related to occlusions and local shape variations by applying approach that delivers a precise analytical solution to integrated positional encoding within the NeRF framework	Avg↓ noted for Exact-NeRF on the Blender dataset in comparison with Mip-NeRF is 0.0242
2024	Lee et al. [147]	×	✓	×	×	A grid-based NeRF to generate sharp novel views from blurry images with reduced training time	Kernel generation time = 0.18 ms
2024	Lin et al. [148]	×	✓	×	×	Leverages random patch sampling and pretrained SR models, it trains up to 23× faster and improves inference speed	Upscale NeRF outputs by 2–4× while maintaining high quality, increasing inference speeds by up to 18× on an NVIDIA V100 GPU and 12.8× on an M1 Pro chip

Table 8 (continued)

Year	Author(s)	EE	PS	MS	HFNVS	Key concept	Performance
2024	Tang et al. [149]	×	✓	×	×	A large-scale dynamic neural rendering system that improves training and rendering speed using a reference decoder and an optical flow & structural similarity-based prior information search method	Training and inference speed on 26 and 1.61–1.35
2024	Chen et al. [157]	×	×	✓	×	CNC framework that optimizes NeRF storage efficiency by leveraging context models that exploit level-wise and dimension-wise dependencies, hash collisions, and occupancy grids	100× and 70× size reduction on Synthetic-NeRF and Tanks and Temples datasets
2025	Luo et al. [163]	×	×	×	✓	A framework for all-in-focus NeRF synthesis using dual-camera systems on smartphones	Average PSNR↑, SSIM↑, and LPIPS↓ on self-collected dataset with foreground-focused and background-focused are 24.31, 0.764, and 0.244 and 24.96, 0.785, and 0.209 respectively
2025	Chen et al. [150]	×	✓	×	×	Introduces density regularization to stabilize scene geometry and depth-based constraints to refine mesh structures without mesh division, ensuring faster convergence	30× – 70× faster training while maintaining high rendering quality
2025	Lee et al. [164]	×	×	×	✓	Quantifies uncertainty in volume density without additional networks, enhancing performance in challenging environments	PSNR↑, SSIM↑, and LPIPS↓ on LLFF dataset (RealF) is 24.97, 0.770, and 0.160 respectively

tially varying blur, achieving high-quality rendering in just 30 min. While vivid colors and fine details are demonstrated, though challenges with motion blur remain, to be addressed. Lee et al. [147] introduced Sharp-NeRF, a grid-based NeRF to generate sharp novel views from blurry images with reduced training time.

Lin et al. [148] suggested a lightweight NeRF+SR pipeline that enhances neural rendering efficiency without complex training or architectural modifications. By leveraging random patch sampling and pretrained SR models, it trains up to 23× faster and improves inference speed by up to 35.7×, making it feasible for consumer devices like MacBook Air M2. This plug-and-play SR strategy maintains high image quality while reducing computational overhead, paving the way for efficient, privacy-preserving NeRF applications on low-power hardware. Tang et al. [149] put forward a large-scale dynamic neural rendering system that improves training speed (by 60%) and rendering speed (by 87%) using a reference decoder and an optical flow & structural similarity-based prior information search method. By leveraging patch-based sampling and prior information from known views, it significantly enhances rendering quality while reducing computational costs, making large-scale autonomous driving simulation more efficient. Chen et al. [150] deliberated ATM-NeRF accelerates NeRF training for mobile rendering by incorporating geometric

constraints, achieving 30× – 70× faster training while maintaining high rendering quality. It introduces density regularization to stabilize scene geometry and depth-based constraints to refine mesh structures without mesh division, ensuring faster convergence. The method produces high-quality, smooth meshes with low storage requirements and supports efficient zoom-in/out operations for mobile applications. While it excels in speed and quality, it lacks relighting generalization and handling sparse input views for enhanced mobile rendering.

3.2.3 Minimized storage requirement

Also, to accelerate NeRF rendering while maintaining quality, Li et al. [151] bring forward an approach leveraging the smooth and continuous nature of viewpoint changes, using feature map with reduced resolution and a compact 2D neural renderer to achieve 30FPS at 1080P image resolution with minimal memory overhead. Zhao et al. [152] introduced TinyNeRF, a three-stage pipeline that reduces the storage demand for voxel radiance fields while preserving speed and synthesis quality by leveraging frequency domain transformation, pruning, and quantization. While, Chen et al. [153] devised bidirectional optical flow NeRF (BOF-NeRF) to address the challenge of estimating correct geometry and synthesizing novel views with NeRF when given fewer input views, utilizing 2D optical flow images to guide NeRF and improve NVS. Takeda et al. [154] used photometric bundle adjustment to train NeRF, allowing it to optimize both camera poses and radiance fields at the same time. This is achieved with sequential data collected from a single monocular camera. To achieve low-cost camera, pose estimation for NeRF, it sparsely estimates camera poses with COLMAP, and the other poses are estimated by interpolation. To enable instant on-device NeRFs, Zhao et al. [155] unveiled Instant-NeRF, a framework that leverages near-memory processing to overcome memory-bound bottlenecks in NeRF training. The proposed solution incorporates a 3D hash mapping function optimized for locality sensitivity, a ray-first point streaming order, and heterogeneous inter-bank parallelism. Collectively, these innovations result in speed improvements of up to 266.1 times compared to SOTA edge GPU benchmarks while preserving rendering quality. Rho et al. [156] propounded a technique to minimize the size of grid-based neural fields used in neural rendering by applying a wavelet transform and a trainable masking approach. This approach achieves decent performance within a 2 MB memory budget by removing over 95% of grid parameters while maintaining visual quality, with the potential for further size reduction in more sophisticated compression pipelines. Additionally, Chen et al. [157] suggested context-based NeRF compression (CNC) framework that optimizes NeRF storage efficiency by leveraging context models that exploit level-wise and dimension-wise dependencies, hash collisions, and occupancy grids. CNC achieves over 100× and 70× size reduction on Synthetic-NeRF and Tanks and Temples datasets, respectively, while enhancing fidelity compared to Instant-NGP [158]. Despite a 1.3× training slowdown, CNC provides a scalable solution for compressing NeRFs, paving the way for efficient deployment in dynamic and large-scale applications.

3.2.4 High-fidelity novel view synthesis

To address the issue of blurry renderings and aliasing artifacts that can occur in certain textures or edges produced by NeRF, Gai et al. [159] developed the edge-guided ray allocation (EGRA-NeRF) module. Additionally, Bian et al. [160] put forward NoPe-NeRF, an

approach that optimizes NeRF and camera poses simultaneously, allowing for improved performance in handling challenging camera trajectories. Neural rendering techniques enable photo-realistic outputs for NVS tasks, but the given approach often face scalability challenges due to per-scene optimization. To tackle issues related to occlusions and local shape variations common in existing NeRF models, Isaac-Medina et al. [161] suggested Exact-NeRF, an approach that delivers a precise analytical solution to integrated positional encoding within the NeRF framework. Exact-NeRF mitigates the frustum approximation issue, achieving accuracy comparable to existing models like Mip-NeRF [162], and extends its applicability to more complex scenarios, including unbounded scenes. Luo et al. [163] unveiled DC-NeRF, a framework for all-in-focus NeRF synthesis using dual-camera systems on smartphones. By leveraging the high resolution of the main camera and the wider depth-of-field of the ultra-wide camera, DC-NeRF aligns and fuses the dual-camera inputs through spatial warping, color matching, and a defocus-aware fusion module. It effectively addresses consistent defocus blur in input views, producing high-quality novel views with adjustable depth-of-field effects. Lee et al. [164] presented Bayesian NeRF that quantifies uncertainty in volume density without additional networks, enhancing performance in challenging environments. Uncertainty modeling is integrated into NeRF, leading to improved rendering quality, particularly for unobserved views, and extending its applicability to non-RGB sensors such as LiDAR and thermal cameras. Significant improvements in simultaneous localization and mapping are demonstrated through the incorporation of occupancy uncertainty, enhancing both mapping and tracking performance. This approach is offered as a scalable and efficient solution for managing uncertainty in real-world 3D scene reconstruction with limited data.

Summary and trend analysis Despite significant progress in NeRF architectural enhancements, key limitations persist that constrain their broader adoption and real-world deployment. Many solutions still rely on high computational resources, specialized hardware, or large memory footprints, which limits scalability on consumer devices and edge platforms. Furthermore, while advances in energy efficiency, processing speed, and storage minimization are notable, they often compromise on rendering fidelity or fail to generalize across diverse and dynamic scenes. Motion blur, sparse input handling, and real-time relighting remain persistent challenges. Future research should focus on developing unified frameworks that balance efficiency, quality, and generalizability. There is also a growing need for adaptive models capable of on-device real-time rendering, robust under occlusions, and scalable across varied environments for applications like autonomous driving, mobile AR/VR, and privacy-preserving metaverse scenarios.

4 Performance evaluation protocols

The efficacy and accuracy of NeRF-based mechanisms are assessed using established evaluation protocols, such as rendering datasets and performance metrics. This section introduces an analysis to assess different methodologies using these protocols.

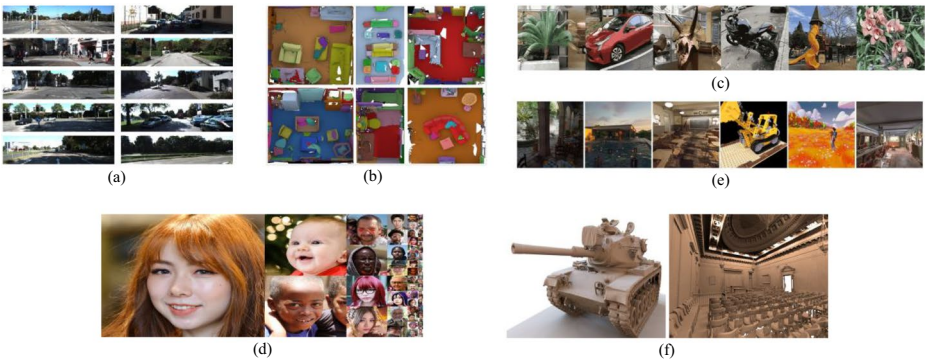


Fig. 11 Sample images from benchmark datasets (**a**) KITTI [165] (**b**) ScanNet [166] (**c**) LLFF [167] (**d**) FFHQ [168] (**e**) Blender [169] (**f**) Tanks and Temples [170]

4.1 Benchmark datasets

A benchmark dataset for NeRF, designed to represent physical identities into the digital realms, is a systematically structured repository of scenes and objects. A sufficiently large corpus is required to accommodate various rendering methodologies, which are essential for thorough evaluation of these algorithms. Figure 11 illustrates few samples of images from the benchmark view synthesis datasets. This section provides an overview of several well-known datasets commonly used in NeRF literature.

The specifics of NeRF datasets are presented in a Table 9, including details such as the number of scans/frames used, and the capturing technologies employed for acquiring images/videos at different resolutions and sizes.

4.2 Performance metrics

Performance metrics are essential for assessing the adequacy of various mechanisms, solutions, or NeRF models against predefined performance criteria, constraints, and trade-offs. Research on rendering techniques indicates that the PSNR, SSIM, and LPIPS are the predominant metrics utilized to evaluate the efficacy of rendering methods. A concise overview of these commonly employed metrics is provided in Table 10.

5 Open research challenges and future scope

The integration of NeRF into metaverse applications necessitates a precise alignment with core technological and experiential requirements that define immersive digital environments. It has significantly advanced photorealistic image rendering and detailed scene reconstruction. However, despite these strides, several challenges hinder their broader adoption. The primary limitations include high rendering quality constraints, slow training and inference speeds, substantial memory requirements, scalability issues, and difficulties in handling dynamic scenes. Furthermore, NeRF models exhibit a lack of generalizability, interpretability, and seamless integration with conventional computer graphics workflows.

Table 9 A brief discussion on benchmark datasets for synthetic view generation

Dataset	Description	Total samples	Capturing device	References
KITTI [165]	A dataset captured from a VW station wagon for use in mobile robotics and autonomous driving research	6 h of traffic scenarios recorded at frequencies ranging from 10 to 100 Hz	Velodyne 3D laser scanner, and a high-precision GPS/IMU inertial navigation system	[23, 87, 110, 120, 123]
RealEstate10K [171]	The RealEstate10K dataset is a large-scale collection of video clips extracted from real estate listings, primarily sourced from YouTube	10 million frames from 80,000 video segments across 10,000 YouTube videos, with high-resolution RGB-D scans requiring about 300 images each	Narrow-baseline stereo cameras encompass VR cameras and the now-widespread dual-lens camera phones	[23]
ShapeNet [172]	ShapeNet represents an extensive repository comprising over 3 million 3D CAD models	Encompasses over 3 million models, with 220,000 of these classified into 3,135 categories	3D sensors	[24]
ScanNet [166]	ScanNet is an extensive RGB-D video dataset that provides critical data including 3D camera poses, surface reconstructions, and instance-level semantic segmentation	707 venues and 1,513 scans	iPad Air2 devices	[27, 35, 39, 45, 96, 97, 99, 109, 111, 160]
ArkitScenes [173]	The dataset facilitates a wide-ranging understanding of 3D indoor scenes in real-world settings	1,661 venues and 5,047 scans	Apple LiDAR scanner	[27, 109]
Local Light Field Fusion (LLFF) [167]	Synthesizing novel views from a set of input images captured by a handheld camera in an irregular grid pattern	It captured 24 scenes with each scene containing 20 to 30 images	Handheld cellphones	[28, 33, 41, 88, 90, 94, 164]
Flickr-Faces-HQ (FFHQ) [168]	A high-quality image dataset of human faces, originally created as a benchmark for GAN	70,000 high-quality PNG images at 1024 × 1024 resolution	Sourced from Flickr, automatically aligned and cropped using dlib	[47, 49, 55, 77]
Blender [169]	The dataset includes various 3D settings such as bulldozers, woodlands, educational spaces, and more, providing a practical foundation for real-time rendering and interactive applications	Each scene in this dataset necessitates approximately 300 images, with the processing time for each image averaging around 20 min on a 64-core CPU	Cycles path tracer	[82, 95, 105, 159, 161]
Tanks and Temples [170]	The dataset comprises a diverse array of high-resolution RGB-D scans that document both indoor and outdoor environments	14 different scenes, totaling 147,791 RGB-D frames	LiDAR scanners	[86, 87, 117, 157, 160]
GazeCapture [174]	GazeCapture represents the first extensive dataset for eye-tracking studies	2.5 million frames gathered from more than 1,450 subjects	iPhones and iPads	[51]

Table 10 An analysis of various evaluation metrics used for performance evaluation

Acronym	Metric	Description	Formula	Reference
PSNR↑	Peak-Signal-to-Noise Ratio	PSNR is a widely used metric for assessing image fidelity, calculated as the logarithmic ratio of maximum signal power to noise power	$\mathrm{PSNR} = 10 \cdot \log_{10}\left(\frac{\mathrm{MAX}_I^2}{\mathrm{MSE}}\right)$	[17]22]– [37]41]– [49, 51, 55, 60, 61]65]– [71]73]– [75, 82, 83, 85, 91, 92]94]– [97]99]– [103, 113, 114, 118, 119, 123, 125, 127, 159, 160, 163, 164]
SSIM↑	Structural Similarity Index Measure	SSIM evaluates image similarity by analyzing luminance, contrast, and structural information. SSIM ranges from −1 to 1, with 1 indicating perfect similarity	$\mathrm{SSIM}(x,y) = \left(\frac{2\mu_x\mu_y + C_1}{\mu_x^2 + \mu_y^2 + C_1}\right) \cdot \left(\frac{2\sigma_{xy} + C_2}{\sigma_x^2 + \sigma_y^2 + C_2}\right)$	[17]22]– [37]40]– [45, 48, 49, 51, 53, 60, 61]65]– [71]73]– [75, 82, 92, 94, 96, 97]100]– [103, 114, 119, 123, 125, 126, 159, 160, 163, 164]
LPIPS↓	Learned Perceptual Image Patch Similarity	LPIPS quantifies perceptual similarity between rendered images and ground truth using a pre-trained CNN	$\mathrm{LPIPS}(x,y) = \sum_l w_l \cdot \lVert \varphi_l(x) - \varphi_{l(y)} \rVert_2^2$	[17]22]– [37]41]– [46, 48, 49, 51, 60, 61]65]– [68, 74, 75, 78, 82, 90, 92, 94, 96]100]– [103]112]– [114, 118, 119, 123, 125, 159, 160, 163, 164]
FID↓	Frechet Inception Distance	FID assesses generative models by comparing the feature distributions of real and generated images using the Inception-v3 network	$\mathrm{FID}(x,y) = \lVert \mu_x - \mu_y \rVert^2 + \mathrm{Tr}(\Sigma_x + \Sigma_y - 2(\Sigma_x \Sigma_y)^{\frac{1}{2}})$	[21, 43, 47, 52]55]– [58, 77, 90, 112]
F-score↑	F-score	F-score is defined as the harmonic mean of precision and recall	$\mathrm{F-score} = 2 \times \frac{\mathrm{Precision} \times \mathrm{Recall}}{\mathrm{Precision} + \mathrm{Recall}}$	[79, 115]
IoU↑	Intersection over Union	IoU measures the overlap between the predicted and ground truth regions	$\mathrm{IoU} = \frac{\lvert y_i \cap \widehat{y_i} \rvert}{\lvert y_i \cup \widehat{y_i} \rvert}$	[79]
CLIP-Score↑	Contrastive Language-Image Pretraining Score	CLIPScore measures how well an image aligns with a given text prompt	$\mathrm{CLIPScore} = \cos(\varphi(I), \varphi(T))$	[76, 104, 107, 117]
MSE↓	Mean Squared Error	MSE measures the average squared difference between predicted and actual values	$\mathrm{MSE} = \frac{1}{N}\sum_{i=1}^{N}\left(y_i - \widehat{y_i}\right)^2$	[91, 113]
RMSE↓	Root Mean Squared Error	It is the square root of the MSE and provides a more interpretable measure of error in the same unit as the original data	$\mathrm{RMSE} = \sqrt{\frac{1}{N}\sum_{i=1}^{N}\left(y_i - \widehat{y_i}\right)^2}$	[120, 124]

Addressing these challenges is essential to unlocking NeRF's full potential across various applications, including scene modeling, content rendering, and beyond. Additionally, Table 11 systematically contrasts the specific demands of metaverse applications with the current capabilities and limitations of NeRF-based approaches.

1. **High rendering quality** NeRF struggles with rendering objects, particularly when processing high-resolution input images that contain intricate details [25, 31, 53, 89] such as high-frequency textures, shading, and view-dependent appearances.
   Future direction: To enhance rendering quality, improvements in image processing and fundamental NeRF methodologies are essential. Recent advancements include per-image transient latent [49, 50, 55] and appearance codes, as demonstrated in [26, 32, 48, 91] which enable control over lighting variations and minor scene changes. Additionally, Mip-NeRF [162] has introduced cone tracing techniques that optimize scene geometry sampling, contributing to improved fidelity in rendered outputs. Further research into refining these techniques could enhance the visual accuracy of NeRF-rendered images.

2. **Slow training and inference speeds** One of NeRF's significant constraints is its computational expense, as slow training and inference speeds [46, 51] hinder real-time rendering. A major contributing factor is the need to evaluate neural networks for each ray in a scene [27], which is highly time-consuming. While certain acceleration techniques exist, they often come at the cost of reduced performance.
   Future direction: Research efforts are focused on optimizing NeRF-based models [52, 111] for speed through three major techniques: advanced volume rendering techniques, hybrid scene representations, and explicit scene feature storage. Advanced volume rendering techniques [144, 145], such as empty space skipping (ignoring computations in irrelevant regions) and early ray termination (halting calculations once sufficient information is gathered), significantly reduce computational overhead. Hybrid scene representations [21, 83, 150] separate learned scene features from color and density MLPs, incorporating additional voxel or spatial-tree features to accelerate inference, though at the 25 of increased memory usage. Explicit scene feature storage [30, 86] further enhances speed by directly storing and retrieving scene features, such as voxel grids, eliminating the need for continuous neural network evaluations while trading off memory efficiency. Furthermore, Instant-NGP [158] and IntrinsicNGP [70] has shown promise by utilizing multi-resolution hashed positional encoding, enabling faster training and inference without compromising accuracy. Further advancements in sparse-view training methods and optimized data structures can enhance real-time applications.

3. **High memory requirements** The extensive memory consumption of NeRF arises from the need to store neural network parameters, particularly for large-scale or high-resolution images. Additionally, the requirement for large training datasets plays a pivotal role, as partial observations of a scene often result in blurry reconstructions, limiting the model's effectiveness in scenarios with sparse input views.
   Future direction: Sparse-view [48, 52] and few-shot NeRF models [25, 153], which incorporate 2D/3D feature extraction using pre-trained CNNs and point cloud data, present an avenue for memory optimization. Integrating sparse-view methodologies

Table 11 A comparative mapping of metaverse requirements with NeRF capabilities and current limitations

Metaverse requirement	NeRF capability	Limitation
Real-time rendering	Fast variants like Instant-NGP improve inference speed	Still challenging for complex or dynamic scenes
Immersive 3D environment	High-fidelity view synthesis from sparse views	Lack of real-time interaction and physics integration
Memory efficiency	Sparse/few-shot NeRFs, single-shot methods are memory efficient	High memory usage persists in high-res or large-scale applications
Robustness to input quality	Use of pre-trained features and pose estimation techniques	Sensitive to calibration errors and poor input views
Dynamic scene support	Extensions like D-NeRF model time-varying scenes	Trade-offs in quality and speed; still an emerging area
Personalized avatar modeling	Face/body modeling with pose-conditioned and latent code-based NeRFs	Still challenging to capture full-body motion and identity preservation in real time
Multi-user interaction and collaboration	Can reconstruct shared virtual scenes	Difficult to edit or update scenes collaboratively in real-time
Generalizability to novel scenes	Few-shot generalization and large-scale data training	Scene-specific training; struggles to generalize to unseen environments without retraining
Scalability to large environments	Sparse voxel grids, incremental scene completion methods are useful	Inefficient global representation; high storage and compute requirements
Interpretability and control	Exploratory work on visualizing learned scene components	Limited transparency; hard to debug or control neural rendering outputs
Multi-modal experience (audio, text, semantics)	Initial efforts using audio/text/sketch-augmented NeRF variants	Early-stage research; lacks robust frameworks for multimodal integration in real-time

with fast NeRF models [68] offers the potential for real-time training and deployment on resource-constrained devices. Furthermore, emerging trends such as single-shot [21, 22, 24, 47] or zero-shot [43] NeRF models leveraging image/latent diffusion methods [151], further enhance efficiency by reducing the dependency on extensive input data while maintaining reconstruction quality.

4. **Sensitivity to input image quality** NeRF models heavily rely on accurate depth estimation [81] and precise camera pose calibration, which are crucial for high-quality scene reconstruction [46, 88]. However, acquiring such reliable input data remains challenging, especially in complex environments where traditional methods often fall short.

Future direction: Incorporating robust pose estimation and self-supervised learning techniques [28, 29, 34] can mitigate NeRF's sensitivity to imperfect input data. Enhancing NeRF's ability to operate with fewer and less accurately calibrated images will expand its usability across diverse real-world applications.

5. **Inability to handle dynamic scenes** NeRF is inherently designed for static scenes, making it challenging to model dynamic environments with moving objects [91, 99]. Current efforts to extend NeRF for capturing scene variations over time remain an active area of research, focusing on developing adaptive frameworks that can efficiently handle temporal changes.
   Future direction: Developing novel techniques that enable real-time updates of NeRF representations [62, 84, 100] as scenes change dynamically is crucial. Potential directions include leveraging temporal coherence, motion priors, and hybrid architectures [35, 37] that integrate both neural and geometric representations.

6. **Personalization and avatar integration** Users should be able to personalize avatars and environments dynamically while preserving identity, expression, and realism. Face-oriented NeRFs [46, 52, 53] offer promising results for facial modeling and expression transfer, but integrating full-body avatars with expression, motion, and real-time response remains technically limited.
   Future direction: To enable dynamic, full-body avatars with real-time expression and motion, future work can integrate parametric human body models with NeRF-based implicit representations. Techniques such as pose-conditioned NeRFs, deformation fields, and temporal encoding [69] can be employed to model expressive motion. Additionally, using real-time face/body tracking and latent code optimization (for identity consistency) can support interactive avatar customization in XR environments.

7. **Multi-user adaptability and interaction** Metaverse platforms must support synchronized, real-time interactions between multiple users in shared digital environments. Most NeRF approaches are single-user and scene-centric [22, 25, 26], lacking built-in support for multi-user collaboration or synchronization. Real-time scene updates and distributed rendering for multiple viewpoints remain an open research challenge.
   Future direction: Developing federated or distributed NeRF rendering pipelines can allow multiple users to interact in shared NeRF-based environments. Techniques such as scene partitioning, multi-agent synchronization protocols, and edge/cloud rendering can support simultaneous scene interaction. Integrating neural radiance caching and asynchronous updates will help maintain low-latency, high-consistency experiences across multiple users and devices.

8. **Generalizability to unseen data** NeRF models are largely scene-specific and struggle to generalize to novel environments [25, 27, 67] without extensive retraining. This limits their deployment in tasks requiring adaptability, such as urban reconstruction and NVS.
   Future direction: To improve generalization, data-driven approaches leveraging large-scale multi-view datasets and pre-trained models have shown promise. Techniques such as [35, 109] demonstrate initial success in generalizing from a few input views. Further

advancements should focus on reducing dependency on extensive viewpoint coverage while maintaining high-fidelity reconstructions.

9. **Seamless integration and usage** Unlike traditional 3D scene representations such as meshes and point clouds, NeRF-based neural representations are not yet fully compatible with existing rendering and editing tools. The absence of standardized workflows makes integration into practical applications challenging.
   Future direction: Future developments should focus on enhancing compatibility between NeRF and traditional graphics workflows. Improved hardware acceleration [62] tailored for neural representations can further close the gap between classical and neural rendering techniques.

10. **Scalability constraints** NeRF implementations predominantly focus on single-object or simple-scene reconstructions, limiting their applicability to large-scale, real-world environments with partial observations [121]. Most volumetric neural rendering approaches struggle to generalize to expansive scenes due to the computational cost and storage demands of a global representation.
    Future direction: Advancements in pose estimation [65] and sparse-view techniques can facilitate real-time deployment [31] on memory-constrained platforms. Efficient neural models for large-scale scenes require innovative storage and retrieval mechanisms akin to VoxelHashing for TSDFs. Future research should emphasize methods that enable incremental scene completion and partial content retrieval, addressing scalability concerns.

11. **Interpretability and explainability** A critical challenge in NeRF and deep learning in general is the lack of interpretability in learned representations. Neural network weights are difficult to analyze in relation to target outputs, such as point colors and opacities in a 3D scene [97]. This opacity hinders debugging and limits broader integration with traditional computer graphics pipelines.
    Future direction: Efforts to enhance interpretability should focus on visualization tools [38–40] that make neural representations more comprehensible. Future work could explore integrating classical graphics principles with neural models [71] to improve controllability and scene modification. Localized neural representations may allow more intuitive scene editing, thereby bridging the gap between classical and neural graphics.

12. **Multi-modal learning** NeRF predominantly focuses on visual signals, yet incorporating additional data types such as textual descriptions, semantic annotations, and audio remains an unexplored avenue.
    Future direction: Multi-modal NeRF models capable of synthesizing corresponding novel audio [60], textual descriptions [54, 55], sketch descriptions [58, 104] and semantic segmentation [105] can broaden applications in AR and telepresence. Developing architectures that seamlessly integrate such modalities will enhance NeRF's versatility.

While NeRF has transformed the field of neural rendering, addressing its current limitations will be pivotal for future advancements. Improvements in rendering quality, computational

efficiency, scalability, interpretability, and integration with classical graphics techniques are key areas for research. Additionally, expanding NeRF's capabilities to handle dynamic scenes and multi-modal learning will unlock new possibilities across diverse applications. The most widely adopted metrics for qualitative assessments within the literature includes, PSNR, SSIM and LPIPS. PSNR quantifies reconstruction fidelity, SSIM evaluates perceptual quality based on structural similarities, and LPIPS assesses alignment with human perception. These metrics may not fully capture the nuanced qualities required for immersive metaverse experiences, such as temporal coherence, motion smoothness, and subjective realism. To address these limitations, emerging evaluation techniques—including temporal consistency metrics, perceptual assessments through user studies, and neural-network-driven quality measures—could be explored in future research to develop a more holistic evaluation framework for metaverse-specific applications. However, as research progresses, NeRF is poised to become a foundational component in next-generation visual computing systems.

6 Conclusions

The metaverse represents the next frontier in digital transformation, seamlessly blending digital and analog elements to unlock new opportunities for learning, social interaction, and business, as well as advancing AR research. One standout technique driving this transformation is the NeRF method, which exhibits promising capabilities in efficient and imaginative 3D synthesis while offering control over various scene parameters. By employing a continuous implicit function defined by a neural network, NeRF offers a robust and adaptable representation of 3D structures based on 2D observations and appearance. The field of neural rendering has witnessed rapid progress in recent years, with this report offering an insightful overview of its flourishing research trajectory. Analyzing over 174 papers, we've identified five key categories of techniques, showcasing applications spanning from computer graphics and avatar and character modeling, interactive media development, and beyond. Our scrutiny extends to the integration of radiance fields with other models including GANs, diffusion models, encoders, and decoders, revealing substantial performance improvements through fusion strategies. Structural enhancements to the original NeRF architecture underscore its potential to tackle increasingly intricate application set-ups. However, challenges persist, notably the high computational overhead and interpretability concerns, necessitating future research efforts. Undoubtedly, NeRF has opened new vistas for exploration in computer vision, inspiring researchers across diverse communities to confront some of the most challenging problems. Moving forward, the focus of NeRF research should prioritize enhancing interpretability, achieving real-time rendering by minimizing computational demands, exploring novel application domains, and bolstering scalability and adaptability.

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Declarations

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